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Master Thesis, Data Science and Marketing Analytics

**Payment Methods and Consumer Spending:
Patterns from Real-World European Transaction Data**

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The full R script for data processing and analysis is available at: [GitHub link](#).

Abstract

This thesis examines whether payment methods (cash, card, mobile, bank transfer) are associated with differences in point-of-sale transaction amounts, and whether these associations vary by context, consumer groups, and the availability of payment choice. Building on salience-based behavioural theories (e.g., pain of paying, mental accounting), which posit that less salient instruments reduce psychological friction and may increase spending, the study assesses whether such patterns appear in real-world data without testing mechanisms directly. Using the ECB's 2024 SPACE one-day payment diary (76,294 POS transactions from 33,395 individuals across 18 euro area countries), the analysis applies ordinary least squares, quantile regression, model-based recursive partitioning, and exploratory correspondence analysis.

The results show that, on average, non-cash methods are associated with higher transaction amounts than cash. Bank transfers show the highest amounts, followed by cards. Mobile payments often fall below cards and are sometimes comparable to, or even lower than, cash in certain subgroups. This contradicts salience-based expectations and suggests that habitual use and contextual constraints may play a stronger role in shaping the relationship between payment method and transaction amount. The associations are heterogeneous across both the spending distribution and contextual conditions. Country and transaction location emerge as the strongest moderators, while demographic differences are less pronounced. Differences between transactions with and without payment method choice availability are small, suggesting a limited role for choice availability in shaping the associations.

Although the observational design prevents causal interpretation, the findings provide useful insights for practice. Aligning payment acceptance strategies with local usage patterns may improve customer experience, and recognising contextual variation can help refine marketing approaches.

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1. Introduction

This thesis examines how point-of-sale transaction amounts are associated with the payment method used—cash, card, mobile, or bank transfer—and how these associations vary across transaction contexts, demographic groups, and the availability of payment choice. It also explores non-causal explanations of these patterns.

Previous research suggests that payment methods such as cash, cards, mobile payments, and bank transfers are linked to different spending levels. Although this study does not attempt to identify underlying mechanisms, the literature offers a range of behavioural and contextual explanations for these differences. These theories highlight why such variation may exist and underscore the importance of examining whether it can also be observed in large-scale, real-world data. One of the most widely cited theories is the pain of paying: the emotional discomfort consumers feel when parting with money (Zellermayer, 1996; Reshadi & Fitzgerald, 2023). This discomfort tends to be stronger when the payment is more salient, such as when paying with physical cash. In contrast, digital methods like cards or mobile payments are thought to reduce salience and weaken the psychological link between the act of purchasing and the experience of financial loss. Closely related is the theory of mental accounting, which proposes that people manage their money using internal budgets (Thaler, 1985, 1999). Less salient payment methods may disrupt this tracking process, leading consumers to underestimate or lose sight of how much they are spending.

However, an alternative line of explanation does not rely solely on psychological salience. Habit formation, payment infrastructure, and situational constraints may also shape both the choice of payment method and the amount spent. For instance, cards are often used for larger purchases simply because carrying large sums of cash is impractical, discouraged by store policies, unhygienic, or considered unsafe (Van der Crujsen et al., 2017; Wakamori & Welte, 2017). Moreover, the relative usage and acceptance of each method vary substantially across different settings. In all these cases, differences in spending may reflect contextual necessity or habitual routines rather than reduced salience.

This study does not aim to test competing explanations, but takes an observational approach to assess whether spending differs systematically across payment methods in real-world data, and whether such associations persist across population subgroups within the euro area. While behavioural theories offer potential interpretations, the central question is whether these differences occur at all, and how robust they are across contexts. Therefore, the research question is:

To what extent are different payment methods (cash, cards, mobile payments, and bank transfers) associated with variation in point-of-sale spending amounts in the euro area (2024), and to what extent do these associations vary across transaction contexts, demographic groups, and the availability of payment choice?

This study draws on the 2024 wave of the European Central Bank’s Study on the Payment Attitudes of Consumers in the Euro Area (SPACE). The dataset contains one-day point-of-sale (POS) diaries from 33,395 respondents across 18 euro area countries, covering 76,294 POS transactions. Although the survey does not measure psychological constructs such as emotional discomfort or budgeting strategies, it offers rich behavioural and contextual information on real-world transactions. It includes over 100 variables with a wide range of demographic characteristics. By focusing on observed behaviour rather than self-reported intentions or hypothetical choices, this thesis adopts a data-driven approach to assess whether transaction amounts differ systematically across payment methods and whether those associations vary across contexts and population subgroups.

Methodologically, the thesis uses multiple analytical methods to explore whether there is an association between payment methods and transaction amounts, and whether this association shows heterogeneity. First, to address whether less salient instruments are linked to higher spending on average, ordinary least squares models are estimated. Payment method is entered both as an ordinal salience score to test for a consistent pattern, and as categorical indicators relative to cash to compare methods directly.

Second, to examine heterogeneity, quantile regression is used to see whether associations differ across the spending distribution, while model-based recursive partitioning detects subgroup structures where the association between payment method and transaction amount shifts across contexts or populations. Additional regressions are used to better understand potential differences, and correspondence analysis visualises relative usage profiles of payment instruments across different groups defined by the model-based partitioning, providing descriptive context. Finally, to test whether associations are stronger when consumers can choose their payment method, OLS models with interaction terms are used.

1.1 Relevance

This thesis contributes to the existing literature by examining the relationship between payment methods and consumer spending using large-scale, real-world data. Much of the prior research in this area is based on small-scale laboratory experiments, typically involving university students and constrained or hypothetical decision-making tasks (Raghubir & Srivastava, 2008; Thomas et al., 2011; Zellermayer, 1996). While these studies offer valuable insights into psychological mechanisms such as the pain of paying, they face concerns about external validity. Laboratory settings often lack real incentives, offer limited contextual variation, and may elicit artificial behaviour due to framing effects. As a result, it remains uncertain whether patterns found in controlled environments generalise to everyday behaviour, particularly in the increasingly digital payment landscape (Liu & DeWitte, 2021). In addition, this study recognises that observed associations—such as higher spending with less salient payment methods—do not necessarily imply underlying psychological causes, such as pain of paying or mental accounting. As highlighted by Jiang (2022), such differences may also reflect habit formation, infrastructure availability, or social norms around payment behaviour.

This study addresses these gaps by drawing on behavioural data from the 2024 wave of the Study on the Payment Attitudes of Consumers in the Euro Area (SPACE), which provides a large-scale dataset of real-life point-of-sale transactions across 18 euro area countries. The dataset includes which payment method was used, what alternatives were available, and over 100 demographic and contextual variables. Unlike most earlier studies—which have focused heavily on credit card behaviour in the United States (Whatman, 2021)—the European setting offers broader variation across cash, cards, mobile payments, and bank transfers (European Central Bank, 2024). These payment methods have evolved rapidly in recent years, particularly following the COVID-19 pandemic, which accelerated the adoption of contactless and mobile transactions (Van der Crujisen & Van der Horst, 2019; European Central Bank, 2022). Many earlier studies predate these developments, raising questions about their relevance in today's more digitally integrated context.

The scale and richness of the SPACE data allow for a detailed and policy-relevant analysis of consumer behaviour. In addition to estimating average differences in spending across payment methods, this study examines how these associations vary across demographic and contextual subgroups, as well as by the availability of payment method choice. The inclusion of extensive control variables enables more precise adjustment for confounding factors. Furthermore, the use of model-based recursive partitioning supports a data-driven exploration of heterogeneity, while correspondence analysis provides descriptive insight into relative usage patterns across identified subgroups. Although the analysis is observational and does not aim to establish causal mechanisms, identifying where and for whom spending differences occur can help guide future research on potential behavioural drivers.

Beyond theoretical relevance, the findings also have practical implications for both marketing strategy and public policy. From a marketing perspective, understanding how payment methods relate to consumer spending can inform the design of checkout processes or retail environments and broader customer strategies. If methods like mobile payments or cards are consistently associated with higher transaction amounts, businesses may benefit from increasing their availability or promoting them during checkout. This could reflect psychological mechanisms such as reduced payment salience, or simply a preference for non-cash methods for larger purchases due to convenience or security. Regardless of the cause, aligning payment options with spending behaviour can help firms influence purchasing decisions and improve sales performance. Moreover, if spending patterns differ across payment methods, payment behaviour might offer useful signals for segmentation and targeting. Consumers who regularly use cards for large purchases may respond differently to loyalty programmes than those who prefer mobile payments. As personalised marketing becomes more important, insight into payment habits can help firms tailor offers and payment options to distinct consumer segments in both physical and digital retail settings.

From a policy perspective, the behavioural effects of payment methods raise important considerations for consumer protection and financial wellbeing. As Zellermayer (1996) noted, the pain of paying can serve a protective function by encouraging self-regulation. If seamless, low-salience payment methods reduce that friction, they could increase the risk of impulsive overspending or financial stress—particularly among younger or more vulnerable consumers. As payment technologies become more invisible and embedded in daily life, regulators may wish to consider measures such as age-based restrictions, default limits, or educational campaigns. Public awareness of the behavioural impact of these methods could support more informed financial decision-making and mitigate unintended risks.

1.2 Structure of the Thesis

This thesis is structured as follows. Chapter 2 reviews the relevant literature, outlining the theoretical concepts and introducing the hypotheses. Chapter 3 describes the dataset, details the data transformations applied, and presents key summary statistics. Chapter 4 explains the methodological approach, providing an in-depth description of each method and its application in this study. Chapter 5 presents the empirical results. Finally, Chapter 6 offers a general discussion of the findings, outlines their policy implications, and addresses the study’s limitations before giving suggestions for future research.

2. Literature Review

This chapter reviews literature on how different payment methods relate to spending behaviour. It introduces three key mechanisms and forms the basis for the hypotheses tested in this thesis. Section 2.1 outlines theoretical concepts; Section 2.2 reviews empirical evidence and formulates hypotheses.

2.1 Theoretical Foundations

Mental Accounting

The concept of mental accounting, introduced by Thaler (1985, 1999), challenges the traditional economic view that individuals treat money as fully interchangeable and make decisions based solely on objective utility. Instead, consumers are seen as constructing mental budgets that assign different subjective values to money depending on its source, purpose, and context. This segmentation often leads to behaviour that deviates from rational norms—such as adhering to arbitrary spending categories or responding more strongly to salient prices or formats. Although Thaler’s work does not explicitly address payment methods, these mechanisms are directly relevant for understanding consumer spending in this context. In particular, the dimensions of timing and salience, central to mental accounting, can help explain how different payment methods shape consumer behaviour.

Building on this foundation, Prelec and Loewenstein (1998) explore how the structure and timing of payments shape consumption experiences. Their model introduces the concept of coupling: the extent to which payment is psychologically linked to consumption. Tightly coupled payments, such as cash transactions, increase salience and reinforce the boundaries of mental budgets. In contrast, decoupled methods, such as credit cards or automatic deductions, reduce this link, weakening the perceived cost and increasing the likelihood of overspending (Prelec and Loewenstein, 1998; Soman, 2003). The less tangible or immediate the payment, the less likely it is to activate psychological resistance. This makes mental accounting particularly relevant in understanding how digital and contactless payment systems can reduce financial self-control. When payment feels abstract or distant, consumers may blur or bypass their mental categories, reducing the effectiveness of internal budgeting as a tool for regulating spending.

Pain of Paying

The pain of paying refers to the emotional discomfort that arises when consumers part with money in exchange for goods or services. The concept was first introduced by Zellermyer (1996), who challenged the classical economic assumption that individuals evaluate transactions in purely rational and utility-maximising terms. Instead, he argued that the act of spending triggers an immediate negative psychological response, shaped not only by the amount spent but also by contextual factors such as timing, perceived control, and the visibility of both the payment and the benefit received. Crucially, Zellermyer positioned this pain as a protective mechanism: by introducing discomfort at the point of

purchase, it can act as a form of self-regulation, helping consumers resist impulsive behaviour and limit overspending. While Zellermyer did not primarily focus on payment instruments, he did suggest that the payment method affects how strongly this discomfort is felt. Cash, due to its visibility, was expected to elicit the most pronounced pain, while automatic bank deductions—being less visible and requiring no active effort—would likely eliminate much of this emotional response. He noted that the effect of credit cards may be ambivalent: while they reduce pain at the time of purchase, they could increase discomfort later when the payment is settled.

Recent work by Reshadi and Fitzgerald (2023) has developed this framework further by offering a more detailed psychological account of what constitutes the pain of paying. They define it as a negative affective response that arises when consumers become aware of losing financial resources, emphasising that this awareness is essential for the discomfort to occur. Their review presents money as a scarce and valued resource, where financial outflows trigger a sense of reduced security. Crucially, they argue that the pain of paying consists of both emotional and cognitive elements: on one hand, feelings such as regret or aversion; on the other, mental engagement, such as attentional focus and the perceived visibility of the transaction.

This has made the concept of payment salience central to explaining variation in the pain of paying. Salience refers to how noticeable or attention-demanding a payment is—both physically and mentally. Soman (2003) positioned payment methods along a spectrum from highly salient forms like cash to low-salience options such as digital or automated payments. More salient payments tend to heighten emotional friction, triggering stronger feelings of loss and greater spending resistance. Reshadi and Fitzgerald (2023) show how reduced salience diminishes both the emotional and cognitive intensity of payment, helping explain why contactless methods are often linked to weaker psychological resistance.

Habitual Use and Transaction Constraints

Beyond psychological mechanisms such as the pain of paying and mental accounting, several behavioural and contextual factors also help explain why spending patterns vary across payment methods. While this perspective has received less attention in the literature, some studies highlight the role of habit, convenience, and situational constraints in shaping payment behaviour. These influences often concern not the act of payment itself, but patterns that develop over time through repetition and context. Consumers tend to stick with the same method in familiar situations simply because it has worked for them before (Van der Crujisen et al., 2017; Deutsche Bundesbank, 2015), creating systematic usage patterns that are unrelated to salience or perceived cost.

Convenience and frictionless use are also major drivers of method choice. Arango and Taylor (2009) show that consumers consider factors such as merchant acceptance, transaction speed, security, budgeting ease, access to funds, and fees. Cards are often favoured over cash due to safety concerns and

the burden of handling physical money (Arango & Taylor, 2009; Wakamori & Welte, 2017). These preferences are further shaped by immediate constraints, such as store acceptance or available funds (Van der Crujisen et al., 2017). Some retailers explicitly require card payments for high-value transactions, reinforcing the association between non-cash methods and larger spending amounts (Wakamori & Welte, 2017). In settings with strong digital infrastructure, such as much of the euro area, these practical factors may drive habitual use of low-salience methods, regardless of psychological resistance. Social and normative dynamics further influence consumer preferences. In more formal or higher-value retail environments, card or mobile payments may be perceived as more appropriate (Foster et al., 2021). Soman (1999) also notes that some high-spending consumers are more likely to select methods like credit cards, introducing potential self-selection bias in observed differences rather than behavioural effects.

During the COVID-19 pandemic, payment behaviour shifted rapidly: Jonker et al. (2021) report a sharp increase in contactless payment preference, driven by hygiene concerns, transaction speed, and merchant reluctance to accept cash. Younger and higher-educated consumers were especially responsive. This shift has largely persisted, reinforcing the idea that practical and social influences, rather than psychological mechanisms, continue to shape payment routines. These contextual developments complicate the interpretation of spending differences across payment methods. Habit, infrastructure, and convenience can shape these patterns. Recognising such influences is essential for interpreting real-world data, as they provide plausible alternatives to purely psychological explanations.

2.2 Empirical Evidence and Hypotheses

A substantial body of research, primarily based on laboratory experiments, shows that less salient payment methods are associated with higher consumer spending. Prelec and Simester (2001) found that participants were willing to pay up to 113% more when using credit cards instead of cash—a result they attributed to reduced psychological salience rather than liquidity constraints. Zellermayer (1996) similarly argued that the tactile and visible nature of cash heightens psychological discomfort, reinforcing the emotional friction of parting with money. Soman (1999, 2003) added that more transparent and tangible payment methods like cash tend to promote self-control, whereas digital methods weaken the perceived link between spending and monetary loss. Raghurir and Srivastava (2008) extended these findings to gift cards, which they linked to the decoupling of payment and consumption. In a European context, Runnemark et al. (2015) found that even debit cards led to higher spending than cash, even after controlling for individual and contextual factors. They attributed this to a diminished perception of money outflow in digital formats, which may undermine spending control. Reshadi and Fitzgerald (2023) further support this line of reasoning, arguing that the pain of paying arises only when consumers are consciously aware of monetary loss—an awareness that declines with less tangible payment formats. While all of these studies report consistent behavioural effects, they

generally attribute the observed spending differences to psychological mechanisms—such as payment salience, friction, and mental budgeting—claiming that the pain of paying is a central explanatory factor.

More recently, research has expanded to include mobile and contactless payments. Falk et al. (2016) found that mobile payments are associated with even higher spending levels than cards, indicating that psychological friction continues to decrease as payment methods become more embedded and less visible. Similarly, Jiang (2022) observed that mobile payments lead to higher spending than cash, particularly for price-sensitive products, across both field and lab settings.

However, the universality of salience as an explanatory mechanism has come under scrutiny. Jiang (2022) found that although mobile payments lead to higher spending, this effect is primarily driven by contextual factors such as convenience, multitasking, and social norms, rather than by diminished payment pain. Liu and DeWitte (2021) actually report that while credit cards still slightly increase spending relative to cash, mobile payments, despite being even less salient, do not consistently produce the same premium in their research. They suggest that the psychological effect may have weakened over time, as real-time account tracking and increased digital literacy have reduced the sense of financial detachment. This shift is particularly relevant in the European context, where credit cards are less common and consumers tend to prioritise convenience and technological compatibility (Van der Cruijssen & Van der Horst, 2019). These findings point to a more complex interplay of factors, where psychological mechanisms interact with familiarity, infrastructure, and context.

Although many studies have found that less salient payment methods are associated with higher consumer spending, these findings are primarily based on tightly controlled laboratory settings, often involving small and homogeneous samples. As a result, their external validity is limited. It remains unclear whether such associations persist in real-world environments with diverse consumers and actual spending decisions. Moreover, most prior studies focus on identifying psychological mechanisms such as the pain of paying or mental accounting, which typically require experimental or survey-based evidence. In the absence of such data, this thesis does not aim to test these mechanisms directly, but instead investigates whether similar patterns appear in large-scale observational data. Specifically, it examines whether average transaction amounts systematically differ across payment methods that vary in psychological salience. Following classifications in prior literature (Soman, 2003; Reshadi & Fitzgerald, 2023), salience is conceptualised as a spectrum: bank transfers are considered least salient, followed by mobile methods, cards, and cash as the most salient. Finding a consistent association between lower salience and higher spending would be in line with experimental findings, though it would not constitute evidence of causality. Based on this, the first hypothesis is as follows:

Hypothesis 1a: Transactions conducted using less salient payment methods are, on average, associated with higher point-of-sale spending amounts in euro area countries.

One particularly informative comparison involves mobile payments and cards, which are functionally similar: both typically draw funds from the same account and rely on comparable infrastructure, costs, and security features (Falk et al., 2016). However, mobile payments offer slightly greater convenience—for example, users can pay without reaching for a wallet, authenticate transactions with a fingerprint rather than a PIN, and complete transactions marginally faster. If systematic spending differences still emerge between these two methods, despite their functional similarity, this may point to subtle psychological mechanisms such as salience or ease of use playing a meaningful role. Given their near-identical financial functionality, any observed spending difference between mobile and card payments would be unlikely to result from liquidity or structural constraints, making this a particularly useful comparison for testing salience-related explanations.

While most experimental studies claim that psychological mechanisms such as payment salience and the pain of paying drive differences in spending, relatively little attention has been given to heterogeneity in these effects. Laboratory experiments typically rely on small, homogeneous samples, often university students, under tightly controlled conditions. As a result, they may obscure important variation in how consumers with different characteristics respond to various payment methods. This limits the external validity of their findings and raises the question of whether such mechanisms operate uniformly across real-world contexts and consumer groups. Rather than testing psychological theories directly, this thesis investigates whether the association between payment method and spending varies systematically across consumers and situations.

Recent work suggests that these effects are unlikely to be homogeneous. Jiang (2022), for example, finds that the spending effects of mobile payments vary depending on both individual traits like multitasking tendencies and price attentiveness, and contextual dynamics like transaction speed and prevailing social norms. He argues that real-world behaviour is embedded in complex environments that controlled experiments may fail to capture. Liu and DeWitte (2021) similarly contend that the psychological effects of certain payment methods may have weakened over time, especially among digitally literate consumers who are accustomed to abstract forms of payment. Their findings point to a growing role of situational fit, technological familiarity, and personal experience in shaping payment behaviour. In line with this, van der Cruysen et al. (2017) show that even the same payment method can trigger different behavioural responses depending on whether it aligns with a consumer's habitual routines or budgeting intentions. Similarly, Jonker et al. (2021) document that demographic and socio-economic characteristics—including age, education level, and digital familiarity—influence how consumers adopt and use payment technologies. These studies highlight that payment effects are likely

to depend not only on the method itself but also on who uses it and in what setting. Together, they support a more contextualised and dynamic understanding of how payment methods interact with demographic, psychological, and behavioural factors.

Recognising this variation is especially important when interpreting large-scale observational data, which includes a broader and more representative sample than most experiments. Understanding who is most affected by different payment methods reveals not only whether an average association exists, but also for whom it matters most. Therefore, a complementary hypothesis is proposed:

Hypothesis 1b: The association between payment method and point-of-sale transaction amount is heterogeneous, with differences expected to be shaped primarily by demographic and contextual factors.

Although the pain of paying framework suggests that less salient payment methods are associated with higher spending, consumers are not always free to choose how they pay. The European Central Bank (2024) reports that 24% of euro area consumers were unable to use their preferred payment method at least once in the past month. This decline in universal acceptance is reflected in falling merchant acceptance rates: between 2019 and 2024, cash acceptance dropped from 96% to 88%, and card acceptance from 93% to 85%. The most common reasons for not accepting cash include low consumer demand, handling difficulties, and security concerns. These structural limitations indicate that consumers may use certain payment methods not out of preference, but due to a lack of alternatives. As a result, the observed association between payment method and spending could reflect contextual constraints rather than salience or habit. For instance, high-value purchases might occur in card-only stores, while smaller cash-only outlets dominate lower spending amounts. To disentangle these influences, it is crucial to examine whether the relationship between payment method and spending differs depending on whether consumers had payment choice. Therefore hypothesis two states:

Hypothesis 2: Payment method is more strongly associated with transaction amount in contexts where consumers had freedom of payment choice.

If a significant association is found in the no-choice group, this would suggest that contextual or structural constraints are the primary drivers of spending differences. For instance, card-only policies for high-value purchases could inflate average transaction amounts in those settings. By contrast, a significant association in the choice group would more likely reflect internal processes such as psychological salience, mental budgeting, or established habits, since consumers selected their preferred method for the situation voluntarily. While this analysis cannot directly identify the underlying mechanism, distinguishing between these groups helps assess whether spending patterns reflect behavioural preference or external context.

3. Data & Descriptive Statistics

This chapter introduces the ECB’s SPACE dataset and outlines the steps taken to prepare it for analysis. Section 3.1 describes the dataset, Section 3.2 details the cleaning and structuring procedures, and Section 3.3 presents descriptive statistics.

3.1 The ECB SPACE Dataset

To analyse how payment methods relate to spending behaviour, this thesis uses the 2024 wave of the Study on the Payment Attitudes of Consumers in the Euro Area (SPACE). Conducted by the European Central Bank (ECB), the SPACE dataset captures consumer preferences and behaviours across 18 countries in the euro area. The study aims to monitor how individuals pay in everyday life, track evolving trends in cash and digital payments, and inform both policy and infrastructure development. It covers a wide range of transaction types, including point-of-sale (POS) purchases and online shopping. However, the analysis in this thesis focuses exclusively on POS transactions. While the dataset includes nearly 30,000 online transactions, these lack key contextual variables and payment options, such as the availability of cash payments, which are central to the research question. In contrast, POS transactions offer a richer behavioural context and include over 76,000 observations. Only these were retained for analysis.

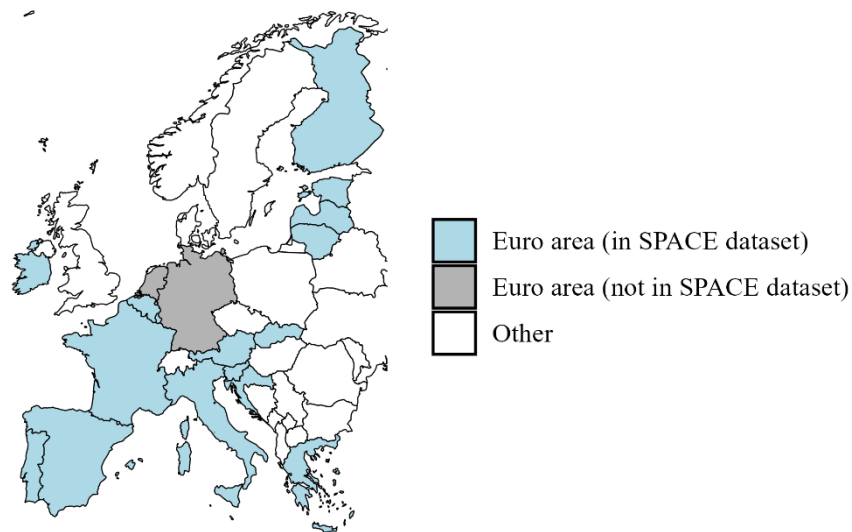


Figure 3.1 Map showing participation in the ECB SPACE 2024 Dataset

Note. This map displays euro area countries that are included in the dataset (in blue). Countries in the euro area that do not participate are grey. All other countries are shown in white. Based on the ECB SPACE 2024 microdata.

While the SPACE dataset includes 18 of the 20 euro area countries, Germany and the Netherlands are excluded (see Figure 3.1). These countries collect comparable data through national surveys managed by their central banks (De Nederlandsche Bank; Deutsche Bundesbank), but methodological differences and strict access procedures prevent direct integration. Given the complexity of securing this data, including formal applications, university collaboration, and on-site access at secure facilities, it falls

outside the scope of this thesis. Relying solely on the ECB-led SPACE dataset allows for high-quality, standardised analysis of consumer payment behaviour across a broad and diverse European sample.

The 2024 SPACE dataset was collected using a combination of computer-assisted telephone and web interviews, with 40,981 respondents across 18 euro area countries. Of these, 33,558 participants reported at least one POS transaction (European Central Bank Annex B, 2024). The sampling approach ensured balanced representation by country, gender, age, and education level, with every adult having a non-zero chance of selection. All countries followed the same standardised questionnaire, translated into the local language to ensure clarity. Data collection involved three phases: a recruitment interview for demographics and consent, a one-day payment diary, and a final survey on attitudes and preferences. To ensure data quality, the ECB applied rigorous cleaning procedures: 0.2% of responses were removed due to inconsistencies, and some missing values were partly imputed using the k-nearest neighbours method. Although the dataset does not report detailed missingness per variable, it is assumed that data quality was handled responsibly. Weighting was also applied to adjust for differences between interview modes and to correct for variation in key behavioural indicators, including cash reserves, initial cash holdings, and POS transaction behaviour, improving the dataset's representativeness and alignment with real-world payment patterns.

The SPACE dataset provides a detailed and structured view of consumer payment behaviour, particularly through its coverage of POS transactions. These transactions are recorded across transaction locations such as stores, restaurants, and entertainment venues. Each transaction includes specific information on the payment method used, enabling analysis of how spending behaviour varies across different formats. This granularity allows for precise assessment of payment method associations. In addition to behavioural records, the dataset contains a wide range of demographics and attitudinal variables that offer insight into consumer preferences, motivations, and concerns. Demographic variables include age, gender, education level, income, and household composition, allowing for subgroup analyses. Respondents report their views on topics such as cash usage, digital payment security, and privacy. These attitudinal measures, combined with rich demographic and transactional data, support a multidimensional analysis of spending behaviour. This makes the dataset particularly well suited for investigating both average associations between payment methods and spending, as well as variation across consumer subgroups and contexts.

3.2 Data Cleaning and Structure

The original SPACE 2024 dataset, published by the ECB, includes responses from 40,981 individuals, of whom 33,558 reported point-of-sale transactions. In total, the original dataset contains over 200 variables, with each respondent able to report up to eight POS transactions. This wide-format structure spreads transaction data across multiple columns (e.g., POS_Amount_1 to POS_Amount_8). To facilitate meaningful analysis of the individual transactions level, which is required by the empirical

strategy, the dataset was reshaped into a long format in which each row represents a single POS transaction. Prior to this transformation, preliminary cleaning steps were performed to improve clarity and usability. These included systematically renaming the original variable labels to ensure they were descriptive and interpretable. Additionally, key steps were taken to address missing values and refine the classification of payment methods. The complete R script for data cleaning, Summary Statistics, and data analytics is available in the associated [GitHub repository](#).

Although the ECB applied initial cleaning procedures, further processing was needed to prepare the SPACE 2024 dataset for analysis. Many variables still contained missing values, often due to response options such as “don’t know” or survey logic that skipped entire modules for respondents to whom the questions or topics did not apply. These non-responses were uniformly recoded as missing values. A preliminary scan confirmed that nearly all rows contained at least one missing entry. To manage this, a threshold-based strategy was applied: variables with over 10% missingness were excluded from the analytical dataset unless essential to the research aims. In total, 71 variables were dropped, with an average structural missingness of 65%. One exception was a key variable central to Hypothesis 2, which captures whether the respondent had access to an alternative payment method at the time of purchase. Originally, this information came from two separate items—one asking if cash was accepted when a non-cash method was used, and the other the reverse. Since each respondent only saw one version based on the payment method used, both variables had over 50% missingness. By combining and recoding them into a single indicator, the missingness was reduced to just 4.1%, making the variable suitable for imputation and inclusion in the final dataset.

For variables with low missingness (under 10%), imputation was used to preserve as much data as possible, particularly for exploratory and heterogeneity analyses. Dropping all rows with any remaining missing values was not considered the most appropriate strategy for this analysis, as it would have resulted in the loss of 33% of all POS transactions. This level of non-random data exclusion could substantially reduce statistical power and risk introducing systematic distortion in the results. While not all imputed variables were used in every model, keeping them available enabled broader testing across subgroups. The full list of retained variables, including those with imputed values, is provided in Appendix D. A total of 45 variables were imputed, with an average missingness of just 1.5%. These low levels of missingness made them suitable for imputation without compromising data quality. In these cases, k-nearest neighbours (KNN) imputation was applied. This technique replaces missing values by examining the k most similar observations in the dataset (here, $k = 5$) and using their values to infer a plausible replacement. KNN offers several advantages over simpler methods like mode or median imputation. Mode imputation assigns the most frequent category to all missing entries, which can distort distributions, reduce variability, and introduce bias, especially when the data is unevenly distributed. By contrast, KNN preserves multivariate relationships and local patterns by drawing on similar observations to estimate missing values. This makes it particularly suitable for behavioural survey data, where

demographic and spending variables are interrelated. As a non-parametric method, KNN requires no assumptions about linearity or distribution, and can flexibly handle both categorical and continuous variables, making it well suited to the complexity of the SPACE dataset.

In addition to handling missing values, the structure of the main explanatory variable—the payment instrument—was refined to enhance theoretical clarity and statistical robustness. The original dataset included nine distinct payment categories which were consolidated into four analytically coherent groups (see Figure 3.2), based on behavioural theory and empirical relevance. Bank cheques, loyalty/gift cards, and “other” were excluded due to limited conceptual alignment with the research question. The resulting four groups are:

- Cash
- Card (physical card(debit and credit) + prepaid card)
- Mobile (mobile or smartwatch payments)
- Bank Transfer (credit transfer + direct debit)

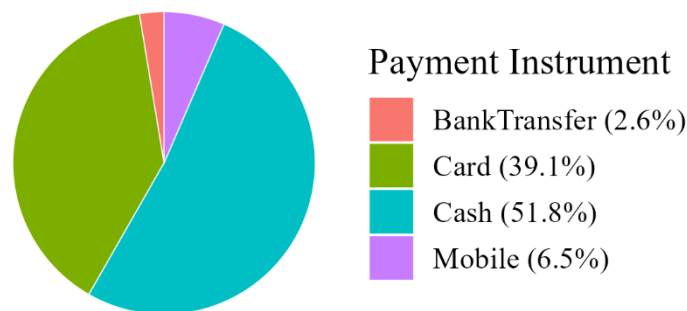


Figure 3.2 Distribution of POS Transactions by Grouped Payment Instrument

Note. This pie chart presents the distribution of POS payment instruments in the ECB SPACE 2024 dataset, after consolidating the original nine payment instruments into four categories. The data are based on 76,294 transactions across 18 euro area countries.

This grouping is grounded in behavioural theory, particularly the concepts of payment salience which varies systematically across instruments and is outlined by Soman (2003). Cash is considered the most salient payment method due to its tangibility and immediate psychological coupling to spending. Cards are slightly less salient but still mentally registered as money. Mobile payments are lower in salience, as they are embedded in devices and require little physical or cognitive effort. Lastly, bank transfers exhibit the lowest salience, often occurring automatically or in the background. Minor distinctions, such as between debit and prepaid cards, are behaviourally negligible and were therefore grouped. Instruments such as loyalty cards, gift cards, and cheques do not feature in the behavioural salience literature and lack a clear conceptual position within the hierarchy. Due to their limited theoretical relevance, they are excluded from the analytical framework. These categories are also relatively rare, with bank cheques (203 observations), loyalty or gift cards (362), and other (910), representing 1.6% of

the data. Retaining them as separate groups would introduce noise and reduce statistical reliability by inflating standard errors and producing unstable estimates.

This four-category framework balances theoretical clarity and statistical robustness. It is applied consistently across all analyses, supporting clearer interpretation and alignment with best practices. After removing observations with payment instruments that are out of scope, the analytical sample consists of 76,294 transactions from 33,395 unique individuals. As shown in Figure 3.2, cash remains the most commonly used payment instrument, accounting for 51.8% of all POS transactions, followed by card payments (39.1%), mobile payments (6.5%), and bank transfers (2.6%). Following these cleaning, restructuring, and imputation steps, the dataset is fully prepared for statistical analysis.

3.3 Summary Statistics

This section presents descriptive statistics and visualisations to explore consumer payment patterns based on the cleaned transaction-level data. The sample comprises 76,294 POS transactions reported by 33,395 unique individuals. Full summary statistics tables are available in Appendix A.

The first descriptive insight is about the distribution of transaction amounts. As shown in Table A.1 (Appendix A) and Figure 3.3 (left), the raw distribution is highly right-skewed: the average transaction is €33.21, while the maximum reaches €35,000. This skew is typical in consumer payment data, where most purchases are low-value, everyday expenses, and only a small number involve large, infrequent purchases. To improve interpretability and facilitate regression modelling, transaction amounts were log-transformed. This transformation also reduces heteroscedasticity and aligns better with the assumptions of linear regression. The resulting distribution, shown in Figure 3.3 (right) on a base-10 logarithmic scale, is approximately normal and indicates that most transactions range between €1 and €100. At the respondent level, Table A.2 (Appendix A) summarises the number of POS transactions recorded during the one-day diary period. On average, individuals reported 2.28 transactions, with the maximum capped at 8, which is the upper limit set by the survey.

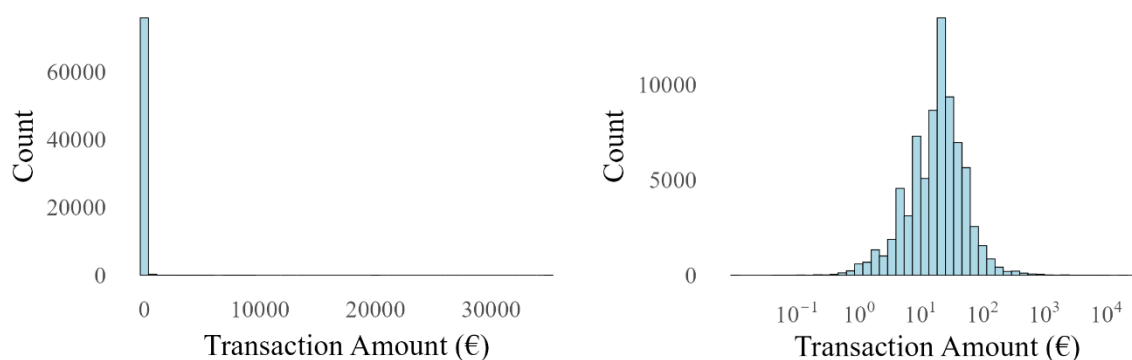


Figure 3.3 Distribution of POS Transaction Amounts (Linear vs. $\log_{10}$ X-Axis)

Note. This dual-panel histogram displays the distribution of POS transaction amounts. The left panel uses a linear x-axis (in euros), while the right panel applies a base-10 logarithmic scale. In both panels, the y-axis indicates the number of transactions per bin. Because the log transformation redistributes values, the y-axis scales differ between the two panels. Based on 76,294 POS transactions from the ECB SPACE 2024 microdata.

In addition to overall transaction amounts, Figure 3.4 shows how spending levels vary across the four grouped payment instruments, using a base-10 logarithmic scale on the x-axis. While the overall shapes are similar, some subtle differences are visible. Cash transactions show a slightly higher concentration in the €1–€10 range, whereas bank transfers display more frequent peaks above €100. The vertical blue lines mark the mean transaction amount per method, which differs moderately across groups: cash averages €27, which is the lowest and bank transfer €70 the highest (see Appendix Table A.3). Standard deviations are substantial across all categories, ranging from €122 for cash to €250 for bank transfers, reflecting considerable variation within each group.

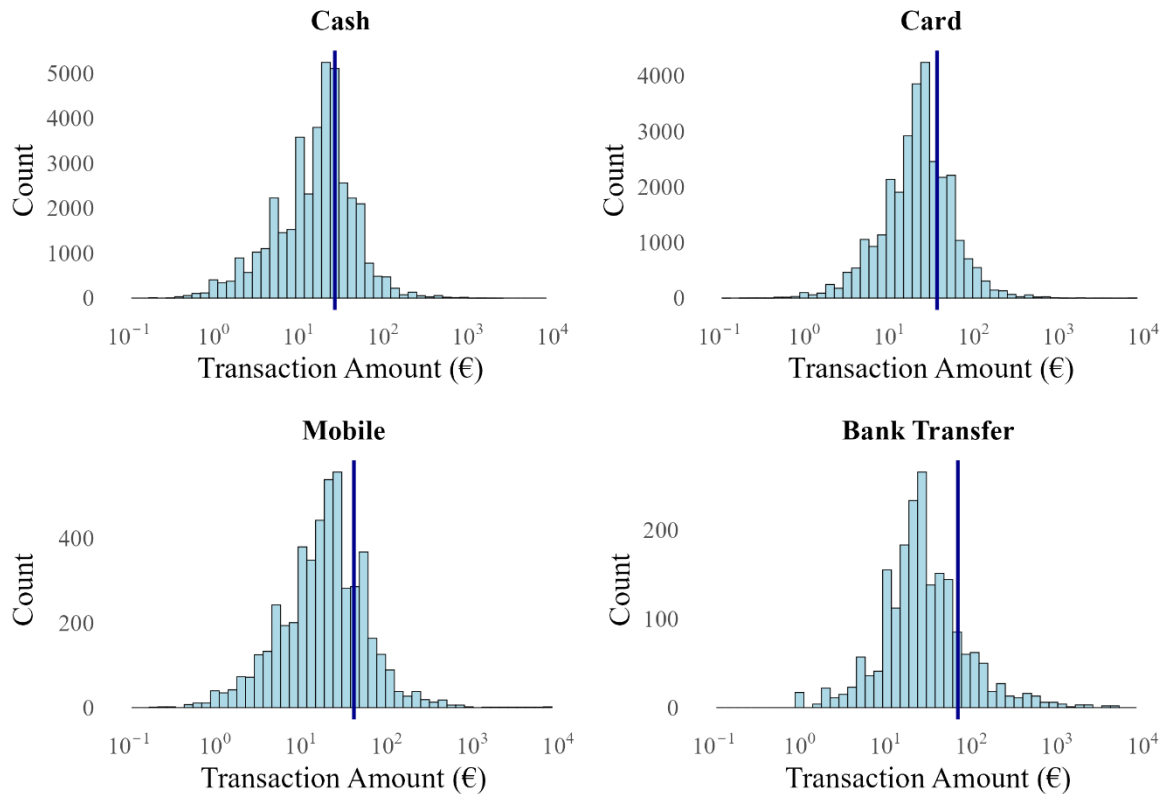


Figure 3.4 Distribution of POS Transaction Amounts by Payment Instrument

Note. This figure shows the distribution of transaction amounts (in euros) by payment instrument. The x-axis is plotted on a base-10 logarithmic scale, and the y-axis reflects the number of individual transactions per bin. Y-axis scales differ across panels due to varying transaction volumes per payment method. The dark blue vertical line in each panel indicates the mean transaction amount for that specific payment method. Data are based on 76,294 POS transactions from the ECB SPACE 2024 survey across 18 euro area countries.

Beyond transaction amounts, several patterns emerge from the data. One key dimension is transaction location. The dataset includes 16 location categories, ranging from supermarkets to petrol stations. As shown in Figure 3.5 and Appendix Table A.4, supermarkets dominate the dataset, accounting for nearly 30% of all POS transactions, followed by day-to-day shops and restaurants. In contrast, transactions at public authorities, hotels, and charities are rare. Average transaction values are highest for services inside the home (€64), other physical locations (€72), and especially hotels (€183), where the highest observed transaction in the entire dataset (€35,000) was recorded. The lowest averages are found at vending machines/ticketing (€13), day-to-day shops (€21), and restaurants/cafés (€22).

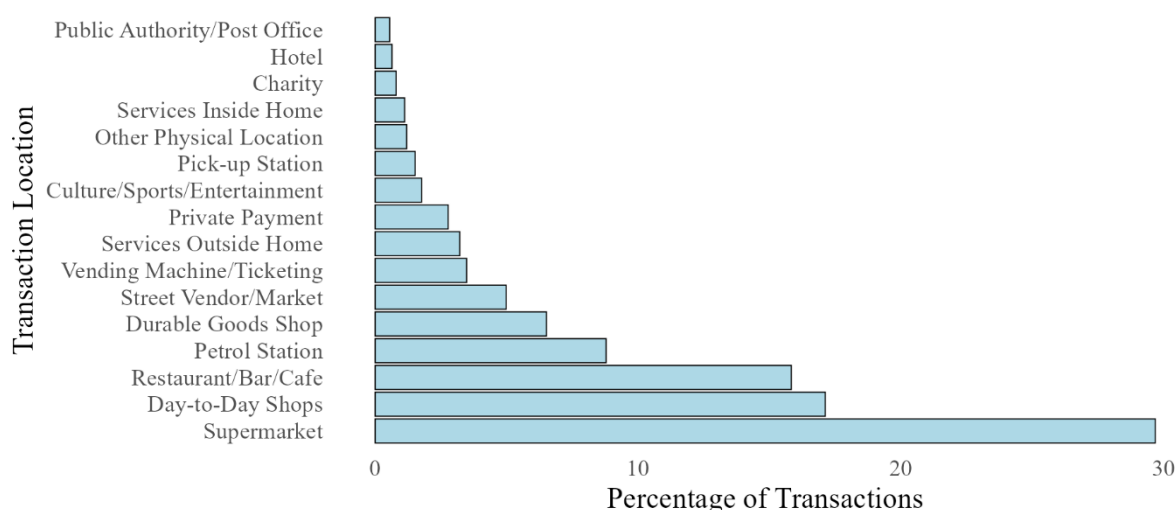


Figure 3.5 Distribution of POS Transactions by Transaction Location

Note. This figure shows the share of POS transactions across 16 different Transaction Locations, based on 76,294 observations from the ECB SPACE 2024 survey across 18 euro area countries.

Cross-country differences are modest but present. As shown in Appendix Table A.5, Luxembourg (€52) and Finland (€41) report the highest average transaction amounts, while Croatia (€24), Slovenia (€26), and Portugal (€26) are at the lower end. These patterns may reflect broader economic conditions such as price levels or income differences. Additionally, age is another interesting factor. Appendix Table A.6 presents the average transaction amount across twelve age categories, which remains relatively stable between €25 and €40. The one notable exception is the 25–29 group, with an average of €42. There are also age differences when looking at payment method choice. Figure 3.6 shows the share of POS transactions by payment instrument across age groups. Mobile payments are more common among younger respondents and decline sharply with age, while the use of cash, cards, and bank transfers remains relatively stable across groups.

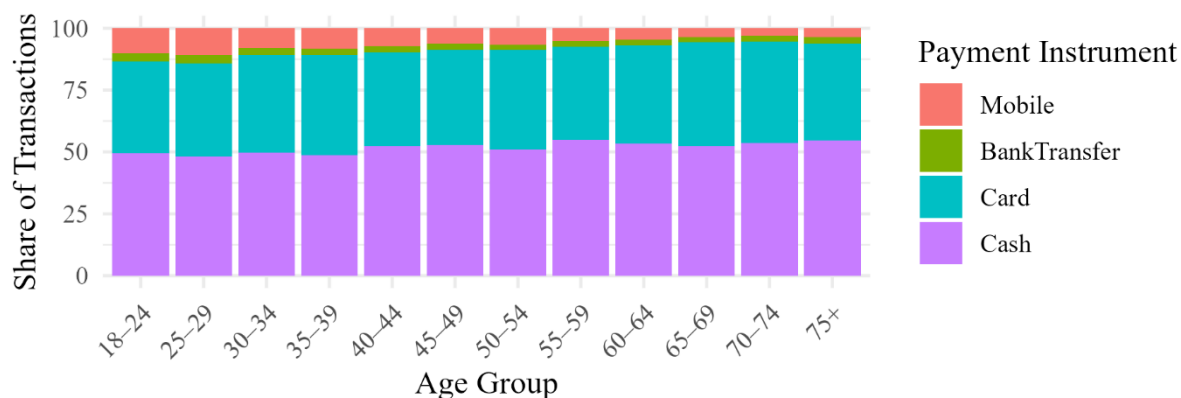


Figure 3.6 Distribution of POS Payment Instruments by Age Group

Note. This stacked bar chart shows the relative share of POS transactions by payment instrument across twelve age groups. Each bar reflects the composition of methods used within that group, expressed as a percentage of total transactions. Based on 76,294 transactions from the ECB SPACE 2024 survey across 18 euro area countries.

While the survey also includes a set of preference questions on why individuals choose certain payment methods, these were not directly analysed within this paper due to limited conceptual clarity, overlap between response options, and inconsistencies in response rates.

4. Methodology

This study investigates to what extent different payment methods are associated with variation in point-of-sale spending amounts in the euro area, and to what extent these associations vary across demographic groups, transaction contexts, and the availability of payment choice. Such variation may result from differences in mental accounting, the pain of paying, or contextual constraints. Based on the salience framework (Soman, 2003; Reshadi & Fitzgerald, 2023), payment methods are expected to differ systematically in associated spending amounts: bank transfers are typically considered the least salient, followed by mobile payments, card payments, and lastly cash.

The analytical approach is observational and hypothesis-testing, combining descriptive statistics with multivariate regression techniques to examine whether observed transaction-level patterns are consistent with salience-based behavioural mechanisms. Specifically, the main analysis estimates ordinary least squares (OLS) regressions on log-transformed transaction amounts, with payment instrument as the primary independent variable. This approach is applied in both Hypothesis 1 and Hypothesis 2. A comprehensive set of control variables is included to adjust for observed confounding. To explore heterogeneity, three complementary methods are applied. First, quantile regression captures distributional differences in the association between payment method and spending, identifying whether effects are stronger for low- or high-value transactions. Second, model-based recursive partitioning (MOB) detects subgroup-specific patterns in these associations, highlighting which demographic or contextual variables drive heterogeneity. Third, correspondence analysis (CA) visualises how the use of different payment methods varies across key demographic or contextual groups, offering an intuitive representation of potential habitual usage patterns. Together, these approaches provide an integrated assessment of both average and heterogeneous effects, without relying on the strong assumptions required for causal identification.

Because the data are observational and consumers self-select into payment methods, causal interpretation is not warranted (Angrist & Pischke, 2009; Imbens & Rubin, 2015). Individuals' payment choices are influenced by transaction context, personal preferences, and habitual patterns, all of which may also affect spending. As a result, the payment instrument variable is likely endogenous, and even with extensive controls, estimates may be affected by omitted variable bias and unobserved heterogeneity (Angrist & Pischke, 2009; Imbens & Rubin, 2015). While endogeneity remains a concern—driven by multiple factors, including self-selection into payment types—the available dataset lacks valid instruments to correct for this. Addressing endogeneity in this context would likely require a large-scale field experiment or detailed follow-up questions in the survey to identify exogenous variation in payment method choice. Furthermore, this study does not aim to predict individual spending behaviour. Predictive modelling often prioritizes statistical fit over theoretical interpretability and is therefore less suited for testing hypotheses about behavioural processes (Shmueli, 2010). Instead, this

analysis examines whether observed spending patterns across payment methods are consistent with theoretical expectations, offering descriptive insights that complement existing experimental findings. The chapter proceeds by outlining the OLS model and its assumptions, followed by extensions using quantile regression, MOB and CA, each then linked to a corresponding hypothesis.

4.1 Analytical Framework: Ordinary Least Squares Regression

This thesis employs OLS regressions as the primary analytical method, prioritising interpretability and theoretical alignment over the predictive complexity of more advanced econometric specifications or “black box” approaches. While techniques such as machine learning can offer high predictive accuracy, it often obscures the role of individual predictors, making it less appropriate for hypothesis-driven behavioural research (Mullainathan & Spiess, 2017). OLS, by contrast, provides transparent estimates, interpretable coefficients, and structured adjustment for confounding variables—qualities essential when the goal is explanation. It enables precise estimation of marginal associations under well-understood assumptions, with outputs that align with theoretical constructs in applied behavioural research. As Wooldridge (2012) notes, linear models remain foundational in empirical economics and the social sciences due to their clarity, flexibility, and testable assumptions. These strengths make OLS particularly well suited for this study.

OLS estimates the relationship between a dependent variable and one or more independent variables by minimising the sum of squared residuals, which is the squared differences between observed and predicted values of the dependent variable (Wooldridge, 2012). Formally, the model solves:

$$\min \sum_{i=1}^n (Y_i - \hat{Y}_i)^2,$$

where Y_i is the observed outcome and $\hat{Y}_i$ is the model’s predicted value for observation i . In linear form, with a log-transformed dependent variable, the model is written as:

$$\log(Y_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + u_i,$$

Where for this thesis $\log(Y_i)$ is the natural logarithm of the transaction amount for observation i . The terms X_{ij} represent the explanatory variables, including both the main independent variable (payment instrument) and a set of control covariates. Each β_j denotes the coefficient representing the marginal association between predictor j and the log-transformed outcome, holding all other variables constant. The error term u_i captures unobserved variations and is assumed to be normally distributed with a mean of zero, constant variance, and independence across observations (Wooldridge, 2012). These assumptions are discussed in more detail in section 4.1.1.

The dependent variable—transaction amount—is highly right-skewed (see Figure 3.4 in Chapter 3.3), with a small number of large transactions inflating the distribution. This skewness can lead to

heteroscedastic residuals, biased standard errors, and excessive influence of outliers, all of which undermine model reliability by potentially increasing the predicted value. To address these issues, a log transformation is applied. This compresses large values, stretches smaller ones, and produces a more symmetric distribution with stabilised variance, thereby improving model fit and aligning with the assumptions of linear regression (Wooldridge, 2012). It also enhances interpretability: in models with log-transformed outcomes, coefficients approximate percentage differences in the outcome per unit change in the predictor, holding all else constant. These values, reported in all result tables for intuitive comparison, are derived from the transformation for coefficients $\hat{\beta}_j$:

$$\% \text{ Change} = (e^{\beta} - 1) \times 100 .$$

To improve the validity of the estimated coefficients, relevant control variables are included to adjust for systematic variation in spending that is unrelated to the payment method. While this does not eliminate omitted variable bias, it reduces its potential impact by accounting for observable confounders and helps prevent subgroup differences from being masked by aggregate effects (Wooldridge, 2012). The specific controls will be discussed later. To further explore heterogeneous effects, interaction terms are introduced in Hypothesis 2 (Angrist & Pischke, 2009). These interaction terms—constructed by multiplying two variables—allow the association between payment method and spending to vary across subgroups, capturing differential sensitivity predicted by behavioural theory. The coefficient on the interaction term reflects how the association between payment method and spending shifts depending on the value of another variable. While such interactions must be theory-driven and used selectively to avoid overfitting, they allow for a more nuanced interpretation of subgroup-specific associations.

To evaluate model fit, R^2 is used to quantify the proportion of variance in the dependent variable explained by the independent variables. It ranges from 0 to 1, with higher values indicating stronger explanatory power. However, R^2 should be interpreted with caution. In data with substantial individual-level variability—such as behavioural or transaction records—a low value does not necessarily reflect a poor model, nor does a high value imply causal validity. Since R^2 always increases with additional predictors, it may overstate performance if not considered alongside theoretical justification. When models are estimated on subgroups, a combined R^2 is calculated across the full sample by comparing actual and predicted values from all submodels. In this study, R^2 is reported for each specification as a comparative measure of explanatory strength.

Lastly, to assess the precision of the estimated coefficients, OLS calculates standard errors, which reflect the expected variability of an estimate across hypothetical repeated samples (Wooldridge, 2012). Formally, the standard error of β_j is the estimated standard deviation of the coefficient's sampling distribution, with smaller values indicating greater precision. Statistical significance is evaluated using

a t-statistic and its associated p-value, testing whether a coefficient differs from zero; the null hypothesis is typically rejected when the p-value falls below 0.05. In smaller samples, low p-values offer useful guidance, as they help distinguish meaningful effects from random noise. However, in large datasets like SPACE, even very small differences in associations can be statistically significant. This is because standard errors shrink with increasing sample size, making it easier to detect even negligible effects. For that reason, interpretation focuses primarily on the size and direction of the coefficients, which offer more substantive insight into behavioural relevance.

4.1.1 Assumptions of Ordinary Least Squares

To ensure the validity of the regression results, the core assumptions underlying OLS estimation are carefully examined. While this study does not aim to establish causality, reliable inference still depends on the accuracy of standard errors and the stability of coefficient estimates. Violations of five key assumptions can lead to invalid conclusions (Wooldridge, 2012). Diagnostic checks are therefore conducted for all regressions (Appendix C), to assess each assumption and maintain statistical integrity.

The first assumption is linearity, which requires that the relationship between each predictor and the dependent variable is linear in parameters—meaning that marginal effects are constant unless interactions are included. Deviations from linearity can produce biased or misleading estimates. Linearity is assessed through residual-versus-fitted plots; systematic curves or patterns may indicate that the assumption does not hold and the need for interaction terms or functional transformations (Wooldridge, 2012). The second assumption is the absence of perfect multicollinearity, which occurs when predictors are highly correlated, making it difficult to distinguish their individual contributions to the dependent variable. This inflates standard errors, undermines statistical power, and can destabilise estimates. Variance Inflation Factors (VIFs) are used to assess this issue, with values above 10 considered problematic (Wooldridge, 2012). In cases of perfect multicollinearity, the model will be specified to automatically drop one of the affected variables to avoid estimation failure.

The third assumption is homoskedasticity, which requires that the error variance remains constant across all levels of the predictors. Heteroscedasticity—when this variance varies—can bias standard errors and distort hypothesis testing. This is checked using residual plots. The log transformation of the dependent variable helps stabilise variance, and clustering standard errors at the respondent level offers additional correction for repeated observations (Wooldridge, 2012; Cameron & Miller, 2015). The fourth assumption concerns the normality of residuals, which is particularly important in small samples. While this is assessed using Q–Q plots, the large sample size in this study allows the Central Limit Theorem to ensure that coefficient estimates remain approximately normally distributed, even if the residuals themselves are not. Minor violations of normality are therefore less concerning (Wooldridge, 2012).

The final assumption is the independence of observations, which requires that each data point is statistically unrelated to the others. This assumption is violated in the current dataset, as respondents can report multiple transactions. With 76,294 observations across 33,395 individuals, the average is just over two transactions per person, introducing intra-individual correlation in the residuals. Ignoring this would underestimate standard errors and potentially overstating statistical significance. To address this, standard errors are clustered at the respondent level using cluster-robust inference (Cameron & Miller, 2015), which allows for arbitrary residual correlation within individuals while assuming independence between them. Although this correction improves inference, it has limitations: the small number of observations per cluster can lead to some instability. Alternatives such as individual-level fixed or random effects are not feasible here due to limited within-person variation and the presence of respondents with only one transaction, which would result in data loss and unreliable estimates. Clustering on respondent ID is therefore the most practical solution. While it does not correct for endogeneity, it yields more robust and potentially more conservative standard errors and supports more robust inference.

4.1.2 Quantile Regression

To complement the OLS analysis and assess heterogeneity, this study also applies quantile regression. This is especially relevant because salience effects may differ across low- and high-value transactions. Introduced by Koenker and Bassett (1978), quantile regression extends classical least squares by estimating conditional quantiles of the dependent variable rather than just the conditional mean. This method is particularly well suited for transaction-level spending data, where the influence of predictors often varies across the distribution. Unlike OLS, which focuses on average marginal effects and assumes homoskedasticity, quantile regression is more robust to skewness, outliers, and non-constant variance—properties commonly observed in behavioural and financial datasets. It also does not assume a specific parametric form for the conditional distribution of the dependent variable.

Diagnostics from Chapter 3.3 (Figure 3.3) reveal substantial right-skewness in transaction amounts, even slightly after log transformation. In such distributions, OLS may overlook important asymmetries in behaviour across spending levels. Quantile regression addresses this by enabling an examination of whether the relationship between payment method and spending differs between low- and high-value transactions. For example, certain methods such as card or mobile may be associated with higher spending, only in the upper tail of the distribution. While this approach does not address endogeneity or establish causality, it enhances the behavioural interpretation by testing whether salience effects vary systematically across transaction sizes.

Formally, the τ -th conditional quantile of spending given covariates $X = x$ is defined as:

$$Q_y(\tau|X = x) = x^\top \beta_\tau$$

where β_τ is the vector of quantile-specific coefficients estimated for a given percentile τ . In this study, the model is estimated for the 25th, 50th (median), and 75th quantiles by assigning asymmetric weights to over- and underpredictions. These cutoffs strike a balance between detail and stability: narrower intervals or extreme tails would yield less reliable estimates due to sparse data, increasing sensitivity to sampling error. These percentiles allow us to assess whether differences in spending are concentrated in the lower, central, or upper segments of the distribution. As with OLS, all quantile models include the same set of control variables to adjust for observable confounding. This allows us to test whether potential behavioural mechanisms, such as salience, systematically vary with transaction size.

For assessing model fit, quantile regression differs substantially from OLS, as it lacks a conventional R^2 statistic. While OLS minimises squared residuals and reports R^2 as the proportion of explained variance, quantile regression minimises absolute deviations, which requires a different approach to model evaluation. Koenker and Machado (1999) propose a natural analog:

$$R^1(\tau) = 1 - \frac{\hat{V}(\tau)}{\tilde{V}(\tau)},$$

where $\hat{V}(\tau)$ is the objective function value of the full model, and $\tilde{V}(\tau)$ is for a restricted model containing only an intercept. This measure compares the model's performance to a naïve benchmark that predicts the same quantile for all observations (Koenker & Machado, 1999). Unlike traditional R^2 , it provides a local measure of relative fit at a specific quantile rather than a global fit across the distribution. This analog can be negative if the model performs worse than the benchmark, underscoring that it should not be interpreted as variance explained. In this study, values of $R^1(\tau)$ are reported for completeness, but interpretation focuses primarily on whether the estimated coefficients are statistically significant, substantively meaningful, and consistent across quantiles.

4.2 Analytical Framework: Model-Based Recursive Partitioning

Model-Based Recursive Partitioning (MOB) is a semi-parametric statistical learning method that integrates classical regression models within a recursive decision tree framework (Zeileis, Hothorn, & Hornik, 2008). Unlike conventional regression, which assumes a single global relationship between predictors and outcomes, MOB helps identify where these relationships vary across subgroups—something especially likely in behavioural contexts. It achieves this by recursively splitting the sample based on covariates and fitting separate parametric models within each resulting node. This enables the detection of structural heterogeneity without the need to pre-specify interaction terms, offering a flexible approach in high-dimensional behavioural data.

Where quantile regression captures when effects differ across the outcome distribution, MOB identifies for whom effects differ based on observed characteristics. For instance, the behavioural salience of a payment method may vary by age (Jonker et al., 2021), psychological traits (Jiang, 2022), or transaction setting (Runnemark et al., 2015). Rather than testing each of these interactions manually, MOB uncovers such subgroup-specific patterns in a data-driven way. Compared to classical tree algorithms such as CART, which assign constant predictions within each node and capture only average outcomes, MOB fits full parametric models within each segment, allowing estimated relationships to vary across subgroups. Compared to more complex models, MOB offers superior interpretability and parameter transparency, attributes essential for testing behavioural hypotheses. MOB balances statistical rigour with theoretical relevance, making it well suited to studies like this one, where uncovering heterogeneity in behavioural responses is central.

At its core, MOB begins with a parametric model $M(Y, \theta)$, where Y is the outcome and θ is a vector of parameters. These parameters are estimated by minimizing an objective function $\psi(Y, \theta)$, which in this study corresponds to the sum of squared residuals from OLS. Formally, this can be written as:

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n \psi(Y_i, \theta).$$

Once the base model is fit, MOB conducts score-based tests to assess whether the estimated parameters are stable across the values of potential partitioning variables. These tests rely on the gradient (first derivative) of the objective function, also known as the score function. If systematic variation is detected, the algorithm identifies the best variable and cut-point to split the sample and refits the model within each resulting subgroup. This process is applied recursively, producing a tree where each terminal node contains its own locally fitted model.

In this study, each terminal node estimates a linear regression with log-transformed transaction amount as the dependent variable and payment method as the key explanatory variable. The set of candidate variables considered for splitting are specified in the following method sections, allowing flexible control over model scope. To ensure reliable estimation and guard against overfitting, the algorithm is configured with a minimum node size of 150 observations and a Bonferroni-adjusted significance level of $\alpha = 0.01$, ensuring that only meaningful splits occur. Additionally, the tree depth is restricted to a maximum of two splits. This conservative setting ensures that only the strongest and most behaviourally meaningful associations are selected for partitioning, enhancing both model stability and interpretability.

To deepen the analysis of the identified splits, two exploratory methods are applied: exploratory OLS regressions to compare coefficient patterns across subgroups, and correspondence analysis to visualise how payment methods relate to key categorical variables selected by the MOB procedure.

4.2.1 Exploratory Regressions

Exploratory regressions are used to explore how associations between payment methods and spending amounts differ across the most behaviourally meaningful MOB-defined subgroups. These models are explicitly exploratory and are not used for statistical inference. The focus lies on the relative size of coefficients, allowing comparisons across subgroups without making formal claims about significance.

This approach is necessary because the MOB procedure selects groups based on where heterogeneity is largest. As Taylor and Tibshirani (2015) explain, applying classical inference after such selection introduces a selective inference problem—commonly referred to as cherry-picking. When regressions are performed on groups that were chosen precisely because they show large differences, standard p-values become biased downward, confidence intervals become too narrow, and apparent effects can be overstated. While correction methods exist, they are complex, not always stable, and not needed for the descriptive purpose of this analysis. Accordingly, no p-values, standard errors, or significance stars are reported. Additionally, no formal significance tests are performed to assess whether the groups identified differ significantly in their associations. However, the combined R^2 values are still presented to provide a general indication of model fit, though they are not interpreted substantively due to potential instability across groups. Model assumptions are also assessed and reported, in order to support transparency and to flag potential issues in model specification or fit.

4.2.2 Correspondence Analysis

To explore whether the use of payment methods differs across relevant groups identified by the MOB analysis, Correspondence Analysis (CA) is applied. It is a multivariate technique for the analysis of categorical data (Greenacre, 2017). CA transforms a contingency table—in this case, cross-tabulating payment method with a categorical variable—into a low-dimensional geometric map in which both row and column categories are represented as points. This allows differences in usage patterns to be visualised across groups. Before applying CA, a chi-squared test of independence is conducted to assess whether a statistically significant association exists between the two categorical variables. This test uses the contingency table to determine whether the distribution of payment methods is independent of categorical groups. A significant test result indicates dependence and justifies the use of CA. In the absence of association, all cells in the table would equal their expected values, and the resulting CA map would show all points clustered at the origin.

CA can be applied to any non-negative data matrix (Groenen & van de Velden, 2004). Here, the matrix with the number of transactions per payment method per category is used, preserving detailed information on relative payment method usage patterns per group and supporting interpretation of the MOB results. This is preferred over alternatives such as CA on average transaction amounts, which would display the mean value per payment method per group. While potentially interesting, this adds

little beyond the exploratory regressions and complicates interpretation. The contingency-table CA therefore provides the clearest and most relevant representation for this analysis.

The analysis follows the methodological framework developed by Greenacre. CA identifies the main axes of variation in categorical associations and projects them geometrically, uncovering patterns that may not be immediately clear from marginal totals. The technique makes no assumptions about underlying distributions or functional forms, and while it does not provide formal statistical inference, it offers valuable exploratory insights. It does so by comparing relative profiles, which are the relative distributions of payment method use within each group, using chi-squared distances. These are derived through singular value decomposition (SVD), which is a matrix factorisation method that reduces the data into a smaller set of latent dimensions, each capturing a distinct share of the overall variation.

To interpret the CA map, two key elements are considered: distance from the origin and directional projection. First, the distance of a category point from the origin reflects how much its profile deviates from the overall average distribution of payment method use. Categories located near the centre indicate usage patterns that are similar to the average distribution, whereas those positioned farther away represent more distinctive or atypical usage. Second, the association between a group and a specific payment method is interpreted through orthogonal projection: this involves drawing a perpendicular line from the group point onto the vector of the payment method. The further this projection lies along the axis—whether in the positive or negative direction—the stronger the relative association. A projection far along the positive direction suggests above-average use, while one in the opposite direction implies below-average use. If the projection lies near the origin, the group's usage of that method is relatively close to the overall average. This geometric interpretation highlights how usage profiles differ between groups and complements model-based methods by uncovering patterns that might otherwise remain hidden.

4.3 Hypothesis 1

To test Hypothesis 1—which consists of two parts—a multi-step analytical strategy is employed. Hypothesis 1a proposes that less salient payment methods are associated with higher average transaction amounts, while Hypothesis 1b posits that this association is heterogeneous, with differences shaped primarily by demographic and contextual factors. The analysis begins with a series of OLS regressions for Hypothesis 1a estimating logged transaction amount as the dependent variable.

To formally test the directional component of Hypothesis 1a, payment method is first entered as an ordinal predictor, reflecting increasing levels of salience, ordered as Bank Transfer, Mobile, Card, and Cash. This specification directly tests whether transaction amounts are systematically associated with the proposed salience gradient. Theoretically, a negative coefficient is expected: more salient payment methods should be associated with lower transaction amounts. In addition, the same models are

estimated using a categorical version of the payment method variable, with Cash as the reference category. While the ordinal model assesses the presence of a linear pattern, the categorical specification quantifies the magnitude of differences between methods without assuming a specific ordering. Both specifications are estimated under three levels of control. The first model includes only the payment method as a predictor and serves as a baseline for evaluating the unadjusted associations.

The second model adds a set of control variables to account for observable heterogeneity in consumer characteristics and transaction contexts. By including a behaviourally motivated set of controls, this model aims to reduce biases from observable factors and improve the interpretability of the estimated association. These control variables include respondent demographics such as age, gender, household size, income group, and employment status, as well as contextual features such as country fixed effects, transaction location, and whether multiple payment options were accepted at the point-of-sale. These controls are theoretically motivated, as both payment method and spending are likely shaped by these factors. For instance, income and employment status influence both a consumer's spending capacity and their access to or preferences for different payment instruments. Crucially, attitudinal or psychological variables are intentionally excluded, as they are conceptually close to the salience mechanism itself and may lead to overcontrol. The second model forms the main specification throughout the remainder of the analyses. The full specification of model 2, expressed in R formula syntax with the names of the categorical variables as used in the data, is given by:

$$\text{Log(Transaction Amount)} \sim \text{Payment Instrument} + \text{Country} + \text{Transaction Location} + \text{Age Group} \\ + \text{Choice Available} + \text{Income Group} + \text{Gender} + \text{Household Size} + \text{Activity Status} .$$

The third model includes the full set of over 100 available covariates, covering attitudinal, behavioural, and contextual dimensions, and is used solely for robustness purposes. It is not intended to identify additional drivers of transaction amount, but to test whether the salience-based association remains consistent under maximal statistical saturation.

To investigate Hypothesis 1b—that the association between payment method and transaction amount is heterogeneous, and to find the sources of this heterogeneity—we employ a complementary, multi-stage approach. The analysis combines techniques that detect the location and structure of heterogeneity with follow-up methods designed to better understand patterns of variation between the identified groups. First, quantile regression is estimated at the 25th, 50th, and 75th percentiles to test whether salience-based associations are stronger in lower or higher segments of the spending distribution. These models use the same controls as the main OLS specification to ensure comparability.

Second, MOB is used to explore whether the associations differ across consumer subgroups. Three MOB specifications are estimated. In MOB Model 1, all available covariates of the data are included as

potential splitting variables. This allows for an unrestricted, data-driven assessment of heterogeneity across the full range of observed characteristics. Model 2 restricts splitting to the same control variables used in the second OLS model, which are country, transaction location, payment choice availability, income, age, gender, household size, and activity status. This version tests for heterogeneity within the core set of observable structural characteristics while maintaining direct comparability to the main regression model. Model 3 focuses only on country and demographic characteristics, including country, age, gender, income group, household size, education, activity status, and access to digital tools like smartphones and online banking. Comparing the resulting trees reveals whether heterogeneity is broadly dispersed across characteristics or concentrated in specific demographic or situational domains.

Finally, the MOB trees are followed by exploratory OLS regressions using the standard controls to compare relative coefficient sizes across subgroups. These regressions omit statistical significance due to the selective inference problem described earlier. CA is then applied to visualise patterns in payment method usage across the key categorical variables identified in the MOB procedure, providing descriptive context to the observed heterogeneity. This layered approach enhances interpretability and supports a deeper understanding of the patterns uncovered through MOB.

4.4 Hypothesis 2

Hypothesis 2 examines whether the association between payment method and transaction amount depends on whether consumers had a choice between cash and non-cash options at the point of sale. The core idea is that behavioural mechanisms like salience are more likely to influence spending when consumers can actively choose how to pay, whereas differences in constrained settings may reflect contextual drivers. To test this, the variable Instrument Choice Available is used, which indicates whether both cash and at least one non-cash method were accepted during the transaction. A value of ‘No’ indicates that the respondent had no alternative option to the method they used.

The empirical strategy uses OLS models with the same dependent variable—the log-transformed transaction amount—and the same set of control variables as in Model 2 of Hypothesis 1a for comparability. Model 1 is estimated on the subset of transactions where both cash and non-cash methods were accepted, capturing situations of genuine behavioural choice. Model 2 restricts the sample to transactions where no choice was available, reflecting constrained payment environments. Model 3 includes all observations and adds an interaction between payment method and instrument choice availability, formally testing whether the spending gap between payment methods differs depending on whether consumers could choose how to pay.

5. Results

This chapter presents the results of the empirical analysis. Section 5.1 addresses Hypothesis 1a, which tests whether average transaction amounts differ systematically across payment methods. Section 5.2 examines Hypothesis 1b, assessing whether the association between payment method and transaction amount is heterogeneous, and whether such differences are shaped primarily by demographic or contextual factors. Section 5.3 addresses Hypothesis 2, which tests whether these associations depend on whether consumers had the freedom to choose their payment instrument.

Before interpreting the regression results, key OLS assumptions are briefly reviewed. Diagnostic tests—including residual and Q-Q plots—are presented in Appendix C (Figures C.1–C.16). Multicollinearity is not a concern: variance inflation factors were tested separately and range from 1.0 to 1.64 across models, well below the conventional threshold of 10. Independence has been addressed in Section 4.1.1 through clustering of standard errors at the respondent level to account for repeated observations. The residual-versus-fitted plots (left panels of Figures C.1–C.16) reveal a mild "bowl-shaped" pattern, suggesting modest deviations from linearity and homoskedasticity, particularly at the distributional extremes. These violations are stable across models and mitigated by clustered standard errors and the large sample size. Q-Q plots (right panels) show moderate deviations from normality in the residual tails, forming a characteristic S-shape. While residual normality is formally violated, the large sample size ensures that inference remains valid due to asymptotic normality under the Central Limit Theorem. Although some assumptions are formally violated, their limited magnitude, consistency across models, and the robustness from large sample size and clustered standard errors support the use of OLS for estimating reliable associational patterns. As the analysis is associational rather than causal, these violations are noted but not considered to undermine validity.

5.1 Hypothesis 1a: Transaction Amounts by Payment Method

Hypothesis 1a examines whether transaction amounts on average are higher when using less salient payment methods—namely, cards, mobile payments, and bank transfers—compared to cash. OLS regressions are estimated in two ways. First, payment method is entered as an ordinal predictor, ordered from least salient (bank transfer) to most salient (cash), to test whether transaction amounts follow the expected salience gradient. Second, a categorical specification is used to assess whether the ordering is consistent across individual payment methods. The diagnostic results (Figures C.1–C.6) show mild assumption violations consistent with those described above.

Table 5.1 presents the ordinal specification. Across all models, coefficients are negative and significant at the 0.1% level, indicating that more salient payment methods are associated with lower average transaction amounts. In Model 1 (no controls), each one-step increase in salience corresponds to an estimated 18% decrease in average transaction amount. Adding standard controls in Model 2 reduces

this to 13%, and the fully adjusted Model 3 to 11%. These reductions suggest that part of the raw association is explained by observable demographic and contextual factors. Overall, the results support a systematic negative relationship between salience and spending.

Table 5.1 OLS Regression Results of Ordered Payment Salience Score on Logged Transaction Amount

	Model 1			Model 2			Model 3		
	Log Transaction Amount			Log Transaction Amount			Log Transaction Amount		
Variables	β	SE	% Δ	β	SE	% Δ	β	SE	% Δ
Ordered salience score	-0.196***	0.007	-17.8	-0.133***	0.006	-12.5	-0.111***	0.009	-10.5
Control variables									
Standard	No			Yes			Yes		
extensive	No			No			Yes		
Constant	3.547***	0.024		3.024***	0.058		3.366***	0.156	
Observations	76,294			76,294			76,294		
R ²	0.018			0.155			0.225		

Note. This table shows results from three OLS regressions estimating the association between ordered payment salience score, from least to most salient, and log-transformed transaction amount. Model 1 includes only payment method; Model 2 adds standard controls (see Section 4.3); Model 3 includes all available controls. Each model includes three columns: β is the regression coefficient, with statistical significance indicated by stars; SE is the standard error, clustered at the respondent level; and % Δ shows the percentage change in transaction amount associated with a one-unit increase in the salience score. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.1$.

In the categorical specification (Table 5.2), all non-cash methods are associated with significantly higher average transaction amounts relative to cash across all three models. In Model 1 (no controls), bank transfers show the largest difference—approximately 79% higher spending than cash—followed by cards (+39%) and mobile payments (+18%). While Table 5.1’s ordinal specification showed a clear negative salience gradient, here the mobile payment coefficient is lower than that for cards, which is inconsistent with the expected salience ordering.

Table 5.2 OLS Regression Results of Payment Method on Logged Transaction Amount

	Model 1			Model 2			Model 3		
	Log Transaction Amount			Log Transaction Amount			Log Transaction Amount		
Variables	β	SE	% Δ	β	SE	% Δ	β	SE	% Δ
Payment Method									
Ref = cash									
Card	0.330***	0.009	39.1	0.239***	0.009	27.0	0.275***	0.008	31.6
Mobile	0.167***	0.021	18.1	0.095***	0.020	10.0	0.197***	0.019	21.8
Bank Transfer	0.582***	0.031	78.9	0.395***	0.029	48.4	0.395***	0.028	48.4
Control variables									
Standard	No			Yes			Yes		
extensive	No			No			Yes		
Constant	2.725***	0.007		2.492***	0.052		2.872***	0.153	
Observations	76,294			76,294			76,294		
R ²	0.026			0.160			0.225		

Note. This table shows results from three OLS regressions estimating the association between payment method and log-transformed transaction amount. Cash is the reference category. Model 1 includes only payment method; Model 2 adds standard controls (see Section 4.3); Model 3 includes all available controls. Each model includes three columns: β is the regression coefficient, with statistical significance indicated by stars; SE is the standard error, clustered at the respondent level; and % Δ indicates the percentage change in transaction amount relative to cash. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.1$.

Adding demographic and contextual controls in Model 2 reduces the magnitudes but keeps the coefficients positive. The spending association for bank transfers drops from +79% to +48%, cards from +39% to +27%, and mobile payments from +18% to +10%. These reductions are expected and likely reflect observable factors such as age or transaction context that influence both payment method choice and spending. In the fully adjusted Model 3, which also includes behavioural and attitudinal variables (see Appendix D), the associations persist. The effect for mobile payments rises to +22% and for card payments to +32%, both higher than in Model 2. This suggests that earlier models may have partially masked behavioural variation. However, many of the additional variables likely proxy for unobserved traits such as digital affinity or habitual payment preferences. Even after extensive adjustment, the estimates may still capture omitted characteristics. The shifts in coefficients across models indicate that omitted variable bias is present in simpler models. This underscores the importance of cautious interpretation when assessing these associations.

Across all model specifications, the relative ranking of payment methods, bank transfer > card > mobile > cash, remains stable. This order is not fully consistent with the salience framework, as mobile payments are consistently associated with lower spending than cards, despite being theoretically less salient. Mobile and card payments are functionally very similar—drawing from the same accounts and involving comparable costs and processes—so any spending difference between them could, in principle, reflect salience effects, although this mechanism cannot be directly tested here. The opposite direction of the effect, however, suggests that salience may not be the primary driver. Instead, these differences may stem from habits, device familiarity, ease of use, or situational cues.

The R^2 values increase from 0.018 to 0.225 in Table 5.1 and from 0.026 to 0.225 in Table 5.2, indicating that payment method alone explains only around 3% of the variation in transaction amounts. Adding contextual and behavioural controls accounts for a larger share, yet the overall explanatory power remains modest. Consumer spending appears highly individualised, shaped by factors not captured in the data. Model 2 of table 5.2, the main specification in this thesis, explains 16% of the variance, leaving 84% unexplained. This large residual component underscores the need for cautious interpretation, as much of the variation likely reflects psychological, situational, or other influences beyond the scope of the available data.

5.2 Hypothesis 1b: Patterns of Heterogeneity in the Association

To test Hypothesis 1b, this section examines whether the association between payment method and transaction amount is heterogeneous and identifies where such heterogeneity occurs. Two complementary methods are applied: quantile regression to assess variation across low-, mid-, and high-value transactions, and MOB to detect subgroup-specific patterns, followed by exploratory regressions and CA to visualise and interpret differences in usage.

Table 5.3 presents the quantile regression results for the 25th, 50th, and 75th percentiles. Card payments are consistently associated with significantly higher spending than cash at all quantiles, but this declines at higher values—falling from +31% at the 25th percentile to +16% at the 75th—suggesting a stronger association for lower-value transactions. In contrast, mobile payments show no significant difference from cash at the 25th percentile, but become positive and significant at the 50th (+4%) and 75th (+11%). Bank transfers display the largest and most stable effects, from +32% to +42%, consistent with earlier findings. The natural analog of R^2 ranges from 0.08 to 0.09, indicating only marginally better performance than a constant prediction at each quantile. The results indicate that payment method associations vary meaningfully across the transaction distribution.

Table 5.3 Quantile Regression Results: Payment Method on Logged Transaction Amount

	Model 1 Q25 Log Transaction Amount			Model 2 Q50 Log Transaction Amount			Model 3 Q75 Log Transaction Amount		
Variables	β	SE	% Δ	β	SE	% Δ	β	SE	% Δ
Payment Method									
Ref = cash									
Card	0.269***	0.010	30.8	0.169***	0.008	18.5	0.144***	0.007	15.5
Mobile	0.028	0.021	2.8	0.037*	0.016	3.7	0.101***	0.016	10.6
Bank Transfer	0.347***	0.032	41.5	0.274***	0.020	31.5	0.308***	0.028	36.1
Control variables									
Standard	Yes			Yes			Yes		
Constant	1.858***	0.082		2.756***	0.050		3.308***	0.037	
Observations	76,294			76,294			76,294		
$R^1(\tau)$	0.092			0.081			0.092		

Note. This table shows results from quantile regressions estimating the association between payment method and log-transformed transaction amount at the 25th, 50th (median), and 75th percentiles. All models include the standard controls (see section 4.3). Each model includes three columns: β is the regression coefficient, with statistical significance indicated by stars; SE is the robust standard error; and % Δ indicates the percentage change in transaction amount relative to cash. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.1$.

Following the quantile regression analysis, the focus shifts to another form of heterogeneity: whether the association varies systematically across consumer groups and contexts, and where this occurs. To examine this, MOB is applied, with the first model allowing splits across all covariates, the second restricting partitioning to the standard controls, and the third limiting partitioning to country and demographic variables. In each tree, the terminal nodes display box plots of logged transaction amounts by payment method. Abbreviations for all labels are provided in Appendix Tables B.1 and B.2.

Figure 5.1 shows the MOB model using the standard control variables, producing the same segmentation as when all available covariates are included (Appendix Figure B.1). This indicates that the standard controls already capture the strongest axes of heterogeneity in how payment method relates to transaction amount. The tree is dominated by transaction location, which drives the top split and one of the second-level splits, the other is by country. This confirms that transaction context is a key moderator. The segmentation yields three location-based subgroups, which are examined in subsequent analysis.

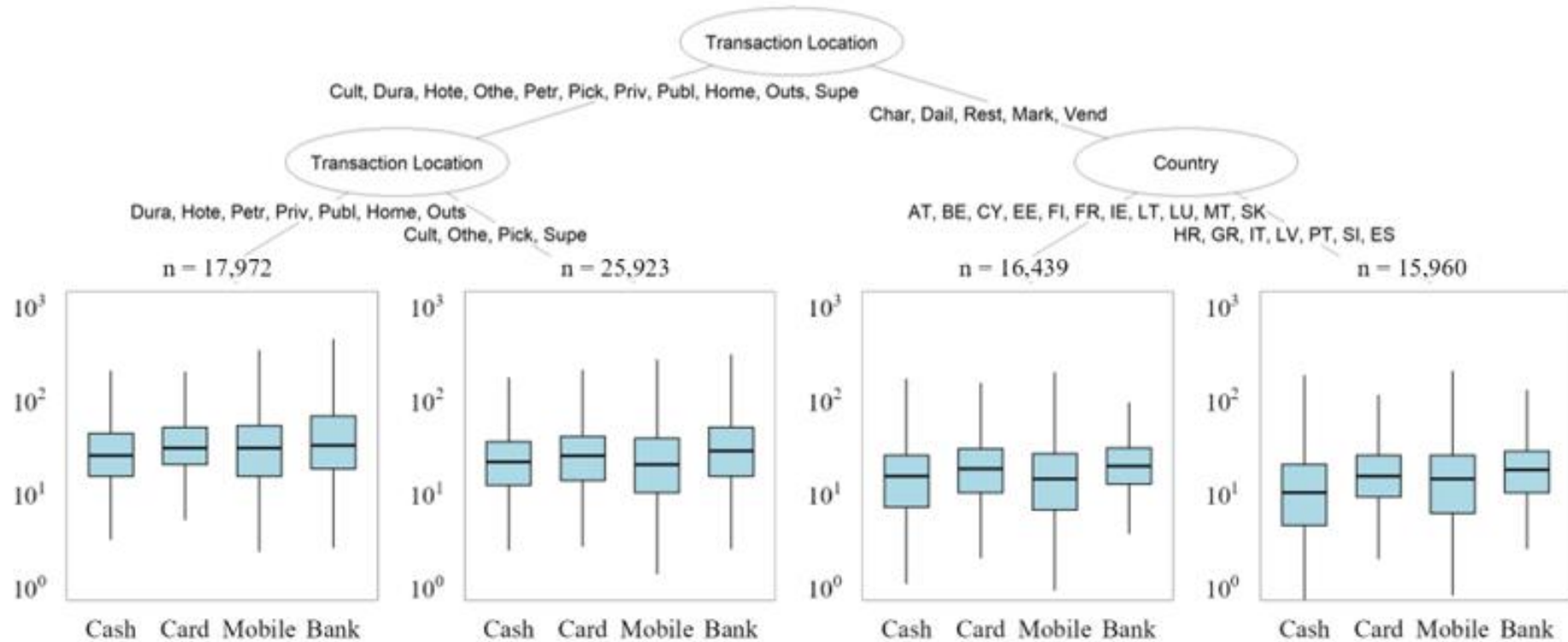


Figure 5.1 MOB Tree Based on Standard Controls (Contextual + Demographic)

Note. This MOB tree is based on the Standard Controls including: payment method, country, transaction location, income group, household size, gender, age group, activity status, and availability of alternative payment methods. Terminal nodes display boxplots of transaction amounts by payment method for each identified subgroup. The y-axis is plotted on a base-10 logarithmic scale, with a reduced display range from 10^0 to 10^3 (instead of the full range from 10^{-1} to 10^4) to improve clarity and focus on the main distribution. Country and location names are abbreviated; full labels are listed in Appendix Tables B1 and B2. For categorical splits, the left branch represents the first-listed category. The analysis is based on 76,294 POS transactions from the ECB SPACE 2024 dataset.

Group 1 (n = 17,972) includes durable goods shops, hotels, petrol stations, private payments, public authorities or post offices, services inside the home, and services outside the home, which are settings typically associated with higher-value or planned purchases. Group 2 (n = 25,923) includes supermarkets, pick-up stations, culture or sports and entertainment venues, and other physical locations, representing functional and routine contexts. Group 3 (n = 32,399) consists of day-to-day shops, restaurants, bars and cafés, street vendors or markets, vending machines or ticketing points, and charitable donations, which are more socially charged and spontaneous settings.

To assess how the association between payment method and transaction amount differs across these transaction location groups, Table 5.4 reports separate OLS regressions for each subgroup. This table does not display standard errors or significance levels due to the selective inference problem. Diagnostic checks (Figures C.7–C.9) show the same assumption patterns as in earlier models. The combined R^2 across all subgroup models is 0.173, compared to 0.160 for the base model in Table 5.2 (Model 2), suggesting that the subgroup structure accounts for only marginal additional variance.

Table 5.4 OLS Regression Results by Transaction Location (MOB Groups)

	Group 1 Log Transaction Amount		Group 2 Log Transaction Amount		Group 3 Log Transaction Amount	
Variables	β	% Δ	β	% Δ	β	% Δ
Payment Method						
Ref = cash						
Card	0.201	22.3	0.121	12.9	0.332	39.4
Mobile	0.120	12.8	-0.046	-4.5	0.169	18.4
Bank Transfer	0.361	43.4	0.320	37.7	0.478	61.3
Control variables						
Standard	Yes		Yes		Yes	
Constant	3.096		2.660		2.459	
Observations	17,972		25,923		32,399	
Combined R^2	0.173		0.173		0.173	

Note. This table shows results from separate OLS regressions estimating the association between payment method and log-transformed transaction amount across three transaction location groups defined via MOB. Group 1 includes: Durable Goods Shop, Hotel, Petrol Station, Private Payment, Public Authority/Post Office, Services Inside Home, Services Outside Home. Group 2 includes: Culture/Sports/Entertainment, Other Physical Location, Pick-up Station, Supermarket. Group 3 includes: Charity, Day-to-Day Shops, Restaurant/Bar/Cafe, Street Vendor/Market, Vending Machine/Ticketing. All models include the standard controls (see Section 4.3). Columns report β , the regression coefficient, and % Δ , the percentage change in average transaction amount relative to cash. Standard errors and significance markers are omitted due to data-driven subgrouping. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries.

In Group 1, which consists primarily of higher-value settings, the associations are moderate and broadly consistent with those in the main specification (Table 5.2). Group 2, which covers more routine environments, shows the smallest range in estimated coefficients, indicating that average spending levels across payment methods are relatively similar. In this group, the estimate for mobile payments is negative, meaning that average spending with mobile is lower than with cash. In contrast, Group 3, which includes more informal or socially charged locations, shows the largest coefficients and the widest spread in associations, suggesting that payment method differences are most pronounced in less

structured, more spontaneous settings. Across all groups, bank transfers are the most stable in their associations, while both mobile and card payments show a wide range of coefficients, with mobile even turning negative in Group 2.

In addition to the regressions, a correspondence analysis was conducted to explore how payment method usage varies by transaction location, and to provide additional insight into the observed heterogeneity in the association between payment method and transaction amount within transaction location groups. Before applying CA, a chi-squared test of independence confirmed a statistically significant association between payment method and transaction location ($p < 1 \times 10^{-16}$), justifying the use of CA.

Figure 5.2 presents the CA map. The first two dimensions together explain 98.3% of the total variation. Dimension 1 (62.9%) captures the primary contrast between more structured, higher-value transaction settings on the right side of the plot and more routine, everyday contexts clustered near the centre. Bank transfers align strongly with the right side, together with locations such as private payments and public authorities—both belonging to Group 1 (pink), which is consistent with this group’s more formal and higher-value nature. Dimension 2 (35.4%) reflects a secondary contrast, separating formal contexts from more informal, socially driven environments such as cafés, vending machines, and street markets, all of which fall within Group 3 (blue).

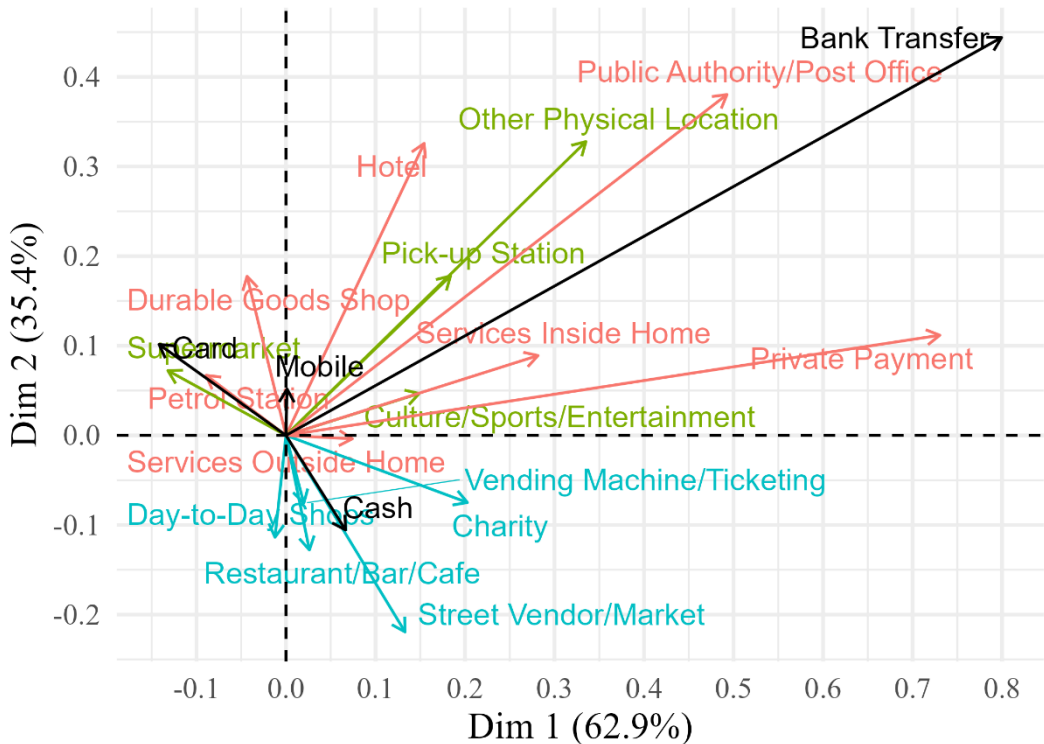


Figure 5.2 Correspondence Analysis Map of Payment Method by Transaction Location
Note. Results from a correspondence analysis of the association between 16 transaction locations and four payment instruments. Transaction locations are color-coded by MOB group (Group 1 = pink, Group 2 = green, Group 3 = blue). Both locations and payment methods, derived from a cross-tabulation, are plotted as points in a shared two-dimensional space. The first two axes (Dim 1 and Dim 2) capture the main latent dimensions of variation. Based on 76,294 POS transactions from the ECB SPACE 2024 survey across 18 euro area countries.

Several transaction locations are positioned notably farther from the origin, indicating payment instrument usage patterns that deviate most from the overall average. Private payments (Group 1, pink) is the most extreme case, projecting strongly on bank transfers, with a clear negative projection on card and a positive projection on cash. This means that, relative to the overall average distribution, private payments involve more bank transfers and, to a lesser extent, more cash use. Public authority/post office (Group 1, pink) is also positioned far from the origin, again projecting strongly on bank transfers but showing slightly below-average use of both card and cash, with a negative projection on those. On the other side of the map, street vendor/market (Group 3, blue) lies far from the origin, along the cash axis at the lower edge of the plot. This indicates above-average cash use. It also projects negatively on both card and mobile, showing that these payment methods are used relatively less in such settings.

Looking at group-level patterns, blue locations (Group 3) cluster predominantly in the lower half of the map and all project positively on cash, reflecting above-average cash use, while projecting negatively on card and mobile, indicating below-average use of these methods. This pattern could potentially be consistent with the regression results, where Group 3 displayed the largest differences in average spending between payment methods. In these settings, certain non-cash payment methods may be less accessible, limiting their use for smaller purchases because the option is unavailable. Card and mobile are used more selectively in these contexts, which could coincide with larger relative gaps in average transaction amounts compared with cash. Group 2 (green) shows a more mixed profile and is, on average, positioned closer to the origin than Group 1. This proximity suggests usage patterns that are closer to the overall average, which could be related to the smaller coefficient spread observed in the regressions. Group 1 (pink) shows a slightly more dispersed pattern than Group 2, with projections pointing in markedly different directions across locations, reflecting internal variation within the group.

Overall, the CA map illustrates that transaction locations differ substantially in their relative payment method usage, with some MOB groups displaying distinct patterns. This first MOB analysis, combining the regression and CA, indicates that there is heterogeneity in the association, occurring in the context of transaction location.

The next MOB model shifts focus away from transaction location, restricting recursive partitioning to demographic and country variables. Figure 5.3 presents the results, with country emerging as the sole splitting variable, underscoring the importance of national context in moderating the association between payment method and transaction amount. Based on this segmentation, four country groups are defined and visualised in Figure 5.4. Group 1 includes Finland and Luxembourg; Group 2 comprises Austria, Belgium, Cyprus, France, Ireland, and Malta; Group 3 includes Estonia, Greece, Italy, Latvia, Lithuania, and Slovakia; and Group 4 consists of Croatia, Portugal, Slovenia, and Spain.

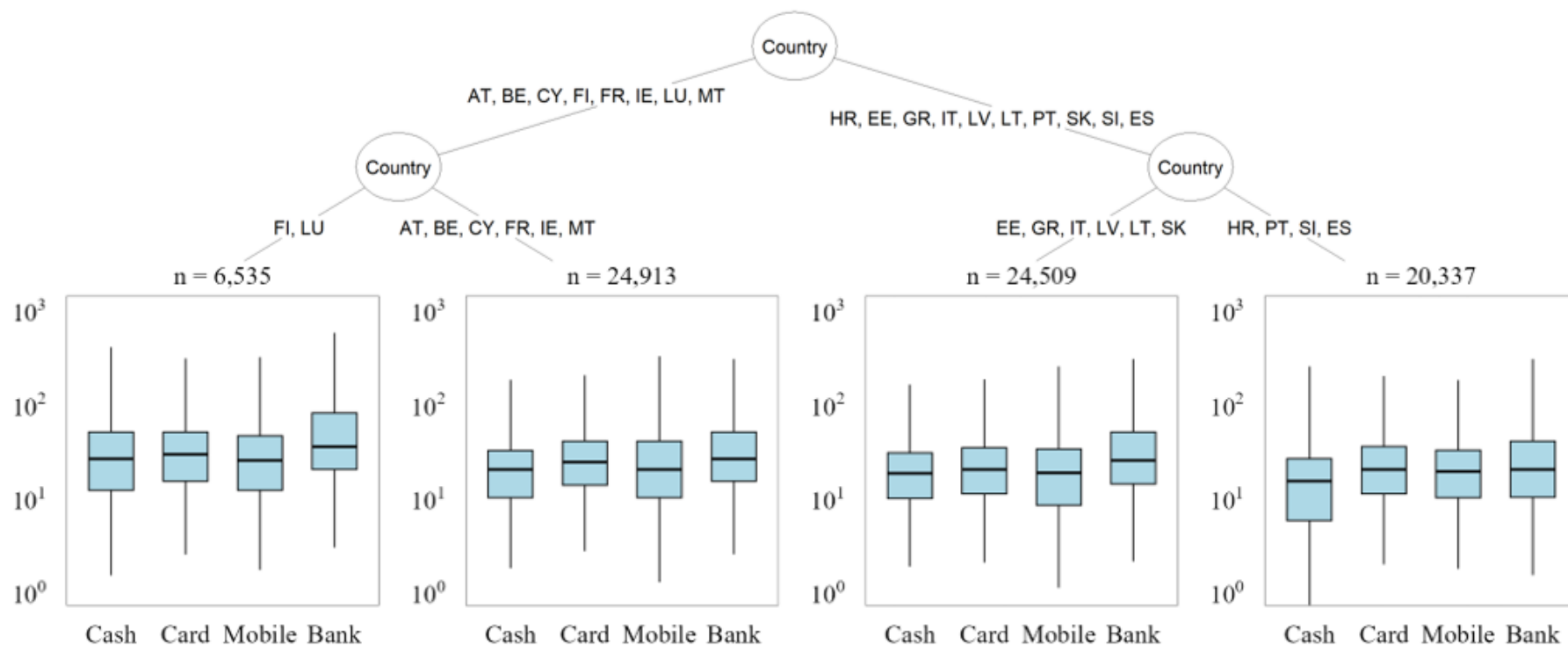


Figure 5.3 MOB Tree with Country and Demographic Variables

Note. This decision tree includes only demographic characteristics and country fixed effects to examine subgroup variation in spending by payment method. Variables considered in the recursive partitioning include country, age, gender, income group, household size, education, activity status, and ownership of a smartphone or online banking account. Terminal nodes display boxplots of transaction amounts by payment method for each identified subgroup. The y-axis is plotted on a base-10 logarithmic scale, with a reduced display range from 10^0 to 10^3 (instead of the full range from 10^{-1} to 10^4) to improve clarity and focus on the main distribution. Abbreviated labels are used for countries; refer to Appendix Table B2 for full names. In categorical splits, the left branch corresponds to the first-listed category. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset.

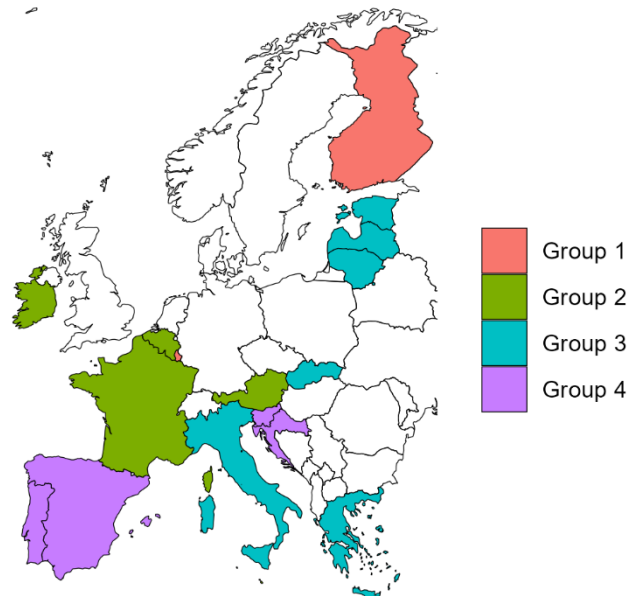


Figure 5.4 Country Groups Derived from MOB Tree Analysis

Note. This map visualizes the four country groups identified through MOB on transaction-level data. Group 1 includes Finland, Luxembourg; Group 2 Austria, Belgium, Cyprus, France, Ireland, Malta; Group 3 includes Estonia, Greece, Italy, Latvia, Lithuania, Slovakia; Group 4 includes Croatia, Portugal, Slovenia, Spain.

Table 5.5 presents OLS results for each of the four country groups, with assumption checks in Figures C.10–C.13 confirming the expected patterns. Group 1 (Finland, Luxembourg) shows relatively small differences across payment methods: mobile payments are negatively associated with transaction amount, and card payments have the lowest positive association of all groups. This points to limited variation in average spending by payment method within this group. Group 2 displays the widest spread of associations and most closely mirrors the full-sample specification in Table 5.1 (Model 2), indicating the strongest differentiation in spending by method. Group 3 lies in between, with mobile showing a small positive association with spending relative to cash and card estimates close to those in group 1.

Group 4 (Croatia, Portugal, Slovenia, Spain) stands out most clearly: mobile payments are associated with 26% higher average transaction amounts relative to cash, substantially higher than in any of the other groups or in the base model (+10%, Table 5.2). Card payments in this group are linked to a 40% higher average transaction amount compared to cash, the largest observed in the analysis. Across all groups, bank transfers show the most consistent positive associations with spending, while mobile remains the most variable. The combined R^2 of 0.170 represents a modest improvement over the base model (0.160), suggesting that while much variation remains unexplained, these country-based subgroups capture some structural differences in how payment methods relate to transaction amount.

Table 5.5 OLS Regression Results by Country (MOB Groups)

	Group 1 Log Transaction Amount		Group 2 Log Transaction Amount		Group 3 Log Transaction Amount		Group 4 Log Transaction Amount	
Variables	β	% Δ	β	% Δ	β	% Δ	β	% Δ
Payment Method								
Ref = cash								
Card	0.116	12.3	0.233	26.2	0.163	17.7	0.334	39.6
Mobile	-0.051	-5.0	0.058	6.0	0.027	2.7	0.231	26.0
Bank Transfer	0.306	35.8	0.404	49.8	0.383	46.6	0.342	40.7
Control variables								
Standard	Yes		Yes		Yes		Yes	
Constant	2.965		2.536		2.141		2.237	
Observations	6,535		24,913		24,509		20,337	
Combined R^2	0.170		0.170		0.170		0.170	

Note. This table reports results from four OLS regressions estimating the association between payment method and log-transformed transaction amount, separately for each group of countries identified via model-based recursive partitioning (MOB). Group 1 includes Finland and Luxembourg; Group 2 includes Austria, Belgium, Cyprus, France, Ireland, and Malta; Group 3 includes Estonia, Greece, Italy, Latvia, Lithuania, and Slovakia; Group 4 includes Croatia, Portugal, Slovenia, and Spain. All models include the standard controls (see Section 4.3). Columns report β , the regression coefficient, and % Δ , the percentage change in average transaction amount relative to cash. Standard errors and significance markers are omitted due to data-driven subgrouping. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries.

After the regression analysis, a CA was conducted to examine the association between country groupings and payment method, providing additional context for the observed heterogeneity. A chi-squared test of independence confirmed a statistically significant relationship between the two variables ($p < 1 \times 10^{-16}$), supporting the use of CA. Figure 5.5 presents the CA map, where the first two dimensions together explain 97.7% of the total variation. Dimension 1 (79.3%) captures the main contrast between countries using relatively more cash (left side) and those using non-cash instruments more than average (right side), with bank transfer and cash positioned nearly opposite each other. Finland and Luxembourg (Group 1, pink) lie furthest to the right, projecting strongly on bank transfers. Group 4 countries (purple) are positioned more towards cash usage. Dimension 2 (18.4%) is primarily driven by mobile payments, which project strongly upward, whereas most other methods are more central or downward. Group 1 (pink) countries project relatively high on mobile, as does Ireland from Group 2 (green). In contrast, several Group 4 (purple) countries project lower on mobile, with Slovenia, Portugal, and Croatia showing negative values.

The country furthest from the origin is Finland (Group 1, pink), indicating the greatest deviation from the average distribution. It projects strongly on bank transfers, and positively on both mobile and card, while projecting strongly negative on cash, which suggests lower-than-average cash use and relatively higher use of the other methods. Luxembourg (Group 1, pink) shows a similar pattern, placing the two Group 1 countries close together in the same direction. This may relate to the regression results for Group 1, which showed the smallest differences in average transaction amounts by method compared to cash, including a negative association for mobile. In such settings, cash may be less prevalent and

alternative methods widely accessible, leading to more uniform usage across contexts. Additionally, Belgium and France (Group 2, green) are also far from the origin, projecting strongly on card and moderately on bank transfers, while projecting negatively on both cash and mobile.

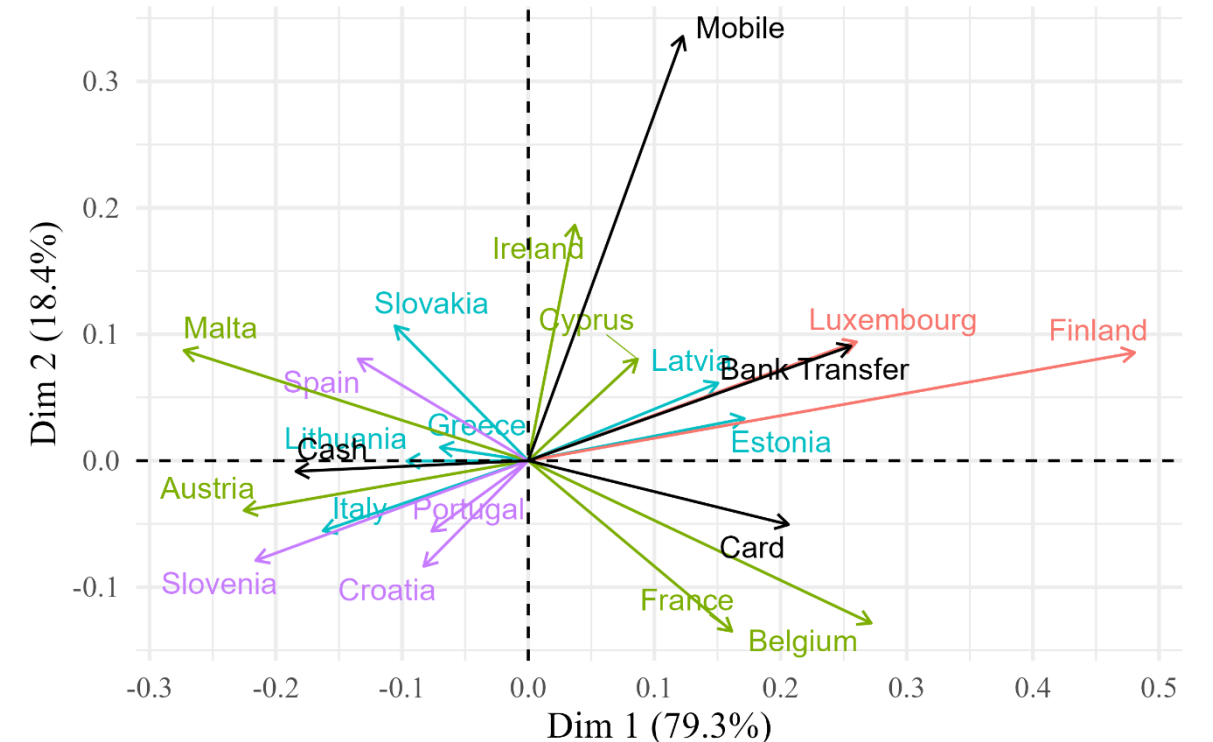


Figure 5.5 Correspondence Analysis Map of Payment Method by Country
Note. Results from a correspondence analysis of the association between 18 euro area countries and four payment instruments. Countries are color-coded by MOB group (Group 1 = pink, Group 2 = green, Group 3 = blue, Group 4 = purple). Both countries and payment methods, derived from a cross-tabulation, are plotted as points in a shared two-dimensional space. The first two axes (Dim 1 and Dim 2) capture the main latent dimensions of variation. Based on 76,294 POS transactions from the ECB SPACE 2024 survey across 18 euro area countries.

Looking at further group patterns, Group 4 (purple) countries are positioned to the left of the map, projecting higher than average on cash. While not exclusively cash-heavy, they generally project lower, and sometimes negative, on other methods. This may help understand the regression results showing the largest differences in average spending between payment methods for this group. One possible factor is that in some countries, such as Croatia, Slovenia, and Spain, certain non-cash methods may be less accessible or less common for lower-value transactions, increasing reliance on cash in these situations and widening spending gaps when other methods are used. Both Groups 2 (green) and 3 (blue) display more mixed and dispersed patterns, with projections varying considerably across payment methods. Group 2 shows greater spread than Group 3 and larger coefficients in the regressions, while Group 3 is positioned closer to the origin, indicating payment method usage patterns nearer to the overall average.

These analyses show that countries differ in their relative use of payment methods, with certain MOB groups displaying particularly distinct patterns. The CA results suggest possible links between usage profiles and the magnitude of spending associations: groups with higher than average cash use tend to

display larger relative gaps between payment methods and transaction amounts in the regressions, whereas groups with more balanced or non-cash-oriented profiles tend to show smaller differences.

Taken together, the quantile regressions, both MOB analyses, and the CAs provide consistent patterns of heterogeneity in the association between payment method and transaction amount, confirming the core proposition of Hypothesis 1b. While Hypothesis 1b anticipated that such variation would be shaped by both demographic and contextual factors, the findings indicate that the strongest moderators are contextual—specifically transaction location and country—both emerging at the top levels of the MOB trees.

5.3 Hypothesis 2: The Role of Available Choice

Hypothesis 2 examines whether the association between payment method and transaction amount differs depending on whether consumers had the freedom to choose how to pay. Table B.3 (Appendix B) shows that 20.9% of cash transactions occurred in contexts where no non-cash option was available, compared to 11.0% of non-cash transactions where cash was unavailable. This asymmetry suggests that cash is more often used in constrained environments.

Table 5.6 presents regression models estimated separately for transactions with and without payment choice, as well as a model with interaction terms. Assumption checks (Figures C.14–C.16) confirm similar patterns as in previous analyses. In Model 1 (choice available), all non-cash methods are associated with significantly higher average transaction amounts relative to cash. Since both cash and non-cash options were available, these differences could reflect behavioural factors such as payment salience or habitual preferences, rather than purely contextual constraints. In Model 2 (no choice), the pattern is broadly similar but with smaller effect sizes; mobile payments are not significantly different from cash. That non-cash methods remain positively associated with spending in constrained contexts may indicate that other structural or contextual factors contribute to these differences. The combined R^2 of Models 1 and 2 is 0.161, only 0.1 percentage point higher than the base model, indicating that the distinction between constrained and unconstrained contexts explains little additional variation of the transaction amounts.

Model 3 introduces interaction terms between payment method and choice availability. On average, having a choice is associated with a +7% higher transaction amount. Based on the interaction terms, the choice group does not differ much from the no-choice group: both bank transfer and mobile show no significant interaction effects. The only significant interaction is for card payments, with a negative coefficient (–9%), indicating that the difference in average spending between card and cash transactions is larger when consumers did not have a choice of payment instrument.

Overall, the analyses do not support Hypothesis 2's expectation that the association between payment method and transaction amount is stronger when consumers have freedom of payment choice. The direction and magnitude of the estimated effects were largely similar in both choice and no-choice contexts, with only a single significant interaction and negligible gains in explanatory power. This suggests that, within the scope of the present data, payment choice availability plays a limited role in moderating these associations.

Table 5.6 OLS Regression Results for Hypothesis 2: Payment Method $\times$ Instrument Choice Availability

	Model 1 Log Transaction Amount			Model 2 Log Transaction Amount			Model 3 Log Transaction Amount		
Variables	β	SE	% Δ	β	SE	% Δ	β	SE	% Δ
Payment Method									
<i>Ref = cash</i>									
Card	0.231***	0.009	26.0	0.260***	0.024	29.7	0.318***	0.023	37.5
Mobile	0.096***	0.021	10.1	0.064	0.056	6.6	0.100	0.056	10.5
Bank Transfer	0.395***	0.032	48.4	0.376***	0.063	45.6	0.407***	0.064	50.2
Choice Available							0.070***	0.014	7.3
Payment Method * Choice Available									
Card * Choice Available							-0.090***	0.024	-8.6
Mobile * Choice Available							-0.007	0.059	-0.7
Bank Transfer * Choice Available							-0.014	0.070	-1.4
Control variables									
Standard	Yes			Yes			Yes		
Constant	2.782***	0.048		2.147***	0.104		2.475***	0.052	
Observations	63,877			12,417			76,294		
Combined R^2	0.161			0.161			0.160		

Note. This table shows OLS regression results for Hypothesis 2, which tests whether the association between payment method and log-transformed transaction amount differs depending on whether both cash and non-cash options were available. Model 1 includes only transactions with payment choice (behavioural context), Model 2 only transactions with no choice (structural constraint), and Model 3 includes an interaction term to test for moderation by choice availability. All models include the standard controls (see section 4.3). Each model includes three columns: β is the regression coefficient, with statistical significance indicated by stars; SE is the standard error, clustered at the respondent level; and % Δ indicates the percentage change in transaction amount relative to cash. Based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries. Standard errors clustered at respondent level. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

6. Conclusion & Discussion

The goal of this study was to examine how different payment methods are associated with variation in point-of-sale transaction amounts, and to assess whether these associations differ across transaction contexts, demographic groups, and the availability of payment choice. This question is grounded in behavioural theories such as the pain of paying (Reshadi & Fitzgerald, 2023), which suggests that less salient payment methods reduce psychological friction and may encourage higher spending, and mental accounting theory (Thaler, 1985; 1999), which posits that low-salience payments can disrupt internal budgeting processes. The study also considers habit formation (Van der Crujisen et al., 2017)—focusing on ease of use, perceived safety, and access to funds—as well as contextual constraints like payment infrastructure and merchant acceptance as alternative explanations for variation in spending behaviour.

To explore whether such patterns emerge outside controlled experimental settings—particularly in the digitally integrated, post-pandemic euro area—this thesis applies an observational approach using data from the 2024 ECB SPACE survey. The dataset comprises one-day spending diaries covering 76,294 POS transactions across 18 euro area countries. The central research question is:

To what extent are different payment methods (cash, cards, mobile payments, and bank transfers) associated with variation in point-of-sale spending amounts in the euro area (2024), and to what extent do these associations vary across transaction contexts, demographic groups, and the availability of payment choice?

The analysis proceeds in three main stages. First, OLS regressions are used to estimate baseline associations and assess their consistency with the salience-based framework. Second, the focus shifts to heterogeneity, applying quantile regression, MOB, and CA to identify and interpret variation across the transaction distribution and between subgroups. Finally, the role of payment choice availability is examined as a potential contextual moderator.

This chapter discusses the findings and their implications in four sections. Section 6.1 presents the main results. Section 6.2 outlines implications for management and public policy. Section 6.3 reflects on limitations, and Section 6.4 offers directions for future research.

6.1 Main Findings and Conceptual Interpretation

Hypothesis 1a states: Transactions conducted using less salient payment methods are, on average, associated with higher point-of-sale spending amounts in euro area countries. Two approaches were used to test this. First, an ordinal specification ranked payment methods from highest to lowest salience (cash = highest, bank transfer = lowest) in OLS models with increasing controls. Coefficients were consistently negative and highly significant, indicating that lower salience is linked to higher transaction

amounts. These results are consistent with Hypothesis 1a. However, the coefficient size declined as more controls were added, and the explained variance was modest, suggesting possible omitted variable bias.

Second, categorical models compared each payment method directly to cash. Across all specifications of hypothesis 1a, non-cash methods were associated with higher spending than cash, following a stable ranking: bank transfer > card > mobile > cash. This broadly aligns with salience-based expectations, except for mobile, which consistently ranked below card. Excluding mobile, the pattern fits well with frameworks such as pain of paying and mental accounting, though the models cannot directly identify the underlying mechanism. Mobile's lower position is consistent across models, despite both mobile and card transactions typically drawing from the same accounts and occurring under similar technical conditions (Falk et al., 2016), and despite mobile being theoretically less salient due to its more frictionless nature. Prior evidence is mixed, with some lab studies (Jiang, 2022) finding mobile effects, whereas Liu & DeWitte (2021) did not, suggesting that the effect may have weakened over time as digital literacy reduced the sense of financial detachment. This study suggest that mobile's salience may operate differently in practice. Possible explanations include limited routine use, trust barriers for high-value purchases, or overlap in psychological cues with cash. Alternatively, usage may reflect habit or context, with mobile preferred for low-value, fast-paced transactions, consistent with Soman's (1999) arguments. In that case, the lower average transaction amounts occur not because mobile payments reduce spending, but because they are more often used for smaller purchases.

Additional results from the Hypothesis 1b analyses provide potential context. The MOB and CA results suggest that when a non-cash method is used less frequently in a country or location, its coefficient relative to cash tends to be larger; when usage is widespread, the spending gap between payment methods narrows. For mobile, this implies that in high-usage environments it may be used interchangeably with cash for everyday purchases, while in low-usage settings—possibly due to acceptance limitations—it may be reserved for larger or more exceptional transactions, which could increase the observed association in those cases. This pattern points more towards contextual and habitual drivers than salience as the primary mechanism behind mobile payment effects.

Overall, Hypothesis 1a is broadly supported. The ordinal salience measure is significant in the expected direction, and the categorical models largely follow the predicted hierarchy. However, the consistent deviation of mobile suggests that psychological salience alone cannot explain the observed associations. Instead, a combination of contextual constraints, habitual use, and behavioural mechanisms is likely at play in the association between payment method and POS transaction amount.

Hypothesis 1b stated: The association between payment method and point-of-sale transaction amount is heterogeneous, with differences expected to be shaped primarily by demographic and contextual factors.

The quantile regressions confirmed variation across the spending distribution: card payments showed the strongest positive association at lower transaction values, while mobile payments became more positively associated at higher quantiles. The MOB analyses revealed pronounced heterogeneity, but mainly along contextual rather than demographic lines. Transaction location and country consistently emerged as top-level splitters, whereas demographic variables such as age or income did not emerge as splitting variables. This only partially aligned with earlier research (Jiang, 2022; Jonker et al., 2021), which emphasised both demographic and contextual variation. The results pointed to structural and situational conditions as stronger moderators of spending differences between payment methods.

The CA maps provided complementary descriptive context by visualising relative payment method use across countries and transaction locations. They showed, for example, that social or informal settings relied more on cash, as did several Southern European countries, while Finland and Luxembourg were strongly oriented towards non-cash methods. Viewed alongside the regression and MOB findings, these profiles suggested that relative usage patterns may be connected to the size of the spending association between payment methods, where higher relative use is linked to a smaller difference in average spending between payment methods, and vice versa. Although CA was descriptive, its consistency with the other analyses reinforced the interpretation that contextual and habitual factors could be key drivers of the association between payment method and transaction amount.

Hypothesis 2 states: Payment method is more strongly associated with transaction amount in contexts where consumers had freedom of payment choice. The findings reveal that average spending patterns across payment methods are largely similar in both constrained and unconstrained contexts. The only notable difference appears for card payments, which are linked to slightly higher transaction amounts in contexts without payment choice. This suggests that even when consumers cannot choose their method of payment, differences in spending across payment types persist—indicating that structural or contextual constraints play some role. At the same time, when choice is available, the associations may also reflect behavioural factors such as habits or psychological mechanisms like mental accounting or the pain of paying. Since most associations remain significant and the added explanatory power from accounting for choice is minimal, the results imply that both contextual and internalised behavioural processes could contribute to shaping payment method related spending patterns. Overall, the results do not support Hypothesis 2: while spending differences between payment methods persist regardless of choice availability, the associations are not stronger when consumers have the freedom to choose.

6.2 Managerial Implications and Policy Relevance

The results consistently show that spending behaviour differs across payment methods, but the size of those differences depends on context. While the exact mechanisms are not proven, the evidence suggests

that habits and local usage patterns influence both payment method choice and the associated spending amounts. This means acceptance strategies should be tailored to the prevailing payment culture rather than applying a one-size-fits-all approach. In cash-heavy environments, including certain countries and more informal settings, retailers should maintain seamless cash handling to meet consumer expectations, while also offering convenient non-cash options for higher-value purchases, where demand for these methods may be stronger. In non-cash-oriented environments, particularly for more expensive purchases, prioritising card and other locally popular non-cash methods with reliable, visible acceptance can help reduce friction at checkout. Aligning acceptance strategies with observed preferences is likely to improve customer satisfaction, even if these patterns are not directly driven by payment salience and will not necessarily increase spending amounts.

At the same time, payment preferences vary per individual. Businesses can use these differences for targeted marketing, customer segmentation, and loyalty programmes. Payment choice is not just a technical step in a transaction—it can act as a behavioural signal of consumer habits and lifestyle, offering valuable insight for both operational decisions and customer engagement strategies. Payment method choice may even function as a proxy for customer profiling when demographic data are limited.

From a policy perspective, the findings offer limited evidence for strong salience-driven behavioural risks like impulsive overspending. However, this does not mean frictionless payments are entirely neutral. Mobile methods, while mostly used for small purchases, may still lead to cumulative overspending if users lose awareness of their total outflows. Deeper investigation, potentially in partnership with banks, could clarify whether spending awareness is compromised and for whom. Given the growing use of mobile payments, policymakers could promote financial literacy around digital payments or, if needed, introduce protective measures, such as restrictions on high-risk tools.

6.3 Limitations and Methodological Reflection

This study benefits from several methodological strengths, including the use of a large and representative dataset covering over 76,000 transactions across 18 euro area countries. It combines multiple analytical techniques guided by clear behavioural hypotheses. These elements enhance the internal consistency and robustness of the findings. However, several limitations must be acknowledged. Because the analysis is based on observational data, it cannot determine whether payment methods causally influence transaction amounts. Reverse relationships are also possible: consumers may self-select a method based on the transaction amount or other factors. Furthermore, unobserved confounders may simultaneously influence both payment choice and spending, introducing endogeneity. In addition, mild violations of OLS assumptions—linearity and homoskedasticity—were detected at distributional extremes. Although mitigated by large sample size and clustered standard errors, these may affect the precision of some estimates.

The study does not directly measure key psychological constructs such as pain of paying, budgeting strategies, or perceived control. Due to this lack of measurement, any interpretation of the results in terms of underlying behavioural mechanisms remains speculative. The observed associations could also reflect other influences than the ones mentioned. These overlapping effects are difficult to disentangle in the current design. The dataset's structure also imposes limitations: each respondent reports only one day of transactions, limiting the ability to observe habitual behaviours or daily variation. Although the sample is representative and diary days were randomly assigned, a longer observation period would be more suitable for capturing within-person dynamics. Moreover, respondents made multiple purchases within that day, and although standard errors are clustered at the respondent level to account for this, the observations are not fully independent. Additionally, a small share of observations included missing values, including the variable on payment choice availability, which were imputed using KNN. While this approach is appropriate under certain conditions, imputation always introduces some uncertainty. Finally, R-squared values across models were relatively low, suggesting that while associations are statistically robust, much of the variation in transaction amounts remains unexplained. These findings should therefore be interpreted with caution.

6.4 Suggestions for Future Research

This study offers large-scale, real-world insights into the association between payment methods and spending, but further research is needed to deepen and refine these findings. A key priority is collecting longer-term transaction data. Multi-day or multi-week spending diaries would allow analysis of within-person variation, showing whether payment methods are associated with differences in both single transactions and cumulative spending. Additionally, to better assess inference, large-scale field experiments in realistic settings with modern digital payment options should be prioritised. Because much existing work relies on student samples or lab environments with limited external validity.

Interpretation would also benefit from direct measurement of psychological constructs such as pain of paying, budgeting strategies, perceived control, and reasons for method choice. This could clarify whether mechanisms like salience or mental accounting are responsible for observed associations. Future studies should also examine how the relative use of a payment method relates to its spending association and testing whether the patterns indicated found in this study are accurate. Additionally, incorporating country-level moderators such as infrastructure or regulation, and expanding the scope beyond the euro area, could help explain cross-national variation.

Finally, newer payment technologies such as Buy Now, Pay Later (BNPL) warrant focused study. These tools may reduce salience and influence spending in distinct ways, particularly among younger users. Given their rapid uptake, BNPL and similar innovations present a timely topic for both academic investigation and policy-oriented assessment.

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8. Appendices

The appendices provide supplementary material, including summary statistics (Appendix A), results tables and figures (Appendix B), and assumption diagnostics (Appendix C)

8.1 Appendix A: Summary Statistics

Table A.1 Descriptive Statistics of Transaction Amounts (in euros)

	Count	Minimum	Median	Mean	Standard Deviation	Maximum
All POS data	76,294	0.01	20.00	33.21	187.08	35,000

Note. This table shows descriptive statistics for POS transaction amounts based on microdata from the ECB SPACE 2024 survey. The sample includes 76,294 transactions across 18 euro area countries and all values are expressed in euros.

Table A.2 Descriptive Statistics of the Number of POS Transactions per Respondent

	Number of Respondents	Minimum	Median	Mean	Standard Deviation	Maximum
All POS data	33,395	1	2.00	2.28	1.42	8

Note. This table shows descriptive statistics for the amount of POS transaction per person based on microdata from the ECB SPACE 2024 survey. The sample includes 33,395 respondents across 18 euro area countries.

Table A.3 Descriptive Statistics of Transaction Amounts per Payment Instrument (in euros)

Payment Instrument	Count	Minimum	Median	Mean	Standard Deviation	Maximum
Cash	39,509	0.01	18	27.01	121.66	20,000.00
Card	29,842	0.10	22	37.02	240.39	35,000.00
Mobile	4,951	0.20	20	41.66	216.13	7,845.00
Bank Transfer	1,992	0.01	25	69.93	250.27	5,139.85

Note. This table shows descriptive statistics for POS transaction amounts per Payment Instrument based on microdata from the ECB SPACE 2024 survey. The sample includes 76,294 transactions across 18 euro area countries and all values are expressed in euros.

Table A.4 Descriptive Statistics of Transaction Amounts per Transaction Location (in euros)

Transaction Location	Count	Minimum	Median	Mean	Standard Deviation	Maximum
Supermarket	22,552	0.01	23.0	33.84	162.37	20,000.00
Day-to-Day Shops	13,186	0.01	15.0	20.65	83.23	8,586.00
Restaurant/Bar/Cafe	12,148	0.07	15.0	21.69	38.13	2,500.00
Petrol Station	6,735	0.05	25.0	34.38	77.32	4,000.00
Durable Goods Shop	4,932	0.10	30.0	49.94	87.17	2,000.00
Street Vendor/Market	3,805	0.10	14.0	27.09	190.90	7,684.00
Vending Machine/Ticketing	2,651	0.20	8.2	12.84	35.97	1,000.00
Services Outside Home	2,450	0.08	26.5	55.27	215.21	7,270.00
Private Payment	2,112	0.01	27.0	64.35	312.48	9,046.00
Culture/Sports/Entertainment	1,324	0.10	20.0	32.14	90.16	2,000.00
Pick-up Station	1,151	0.75	22.0	36.72	143.61	4,499.00
Other Physical Location	896	0.01	24.0	71.83	203.53	3,518.18
Services Inside Home	851	1.00	30.0	64.30	232.26	5,139.85
Charity	609	0.20	19.0	24.20	30.59	446.00
Hotel	477	2.00	50.0	183.28	1,638.96	35,000.00
Public Authority/Post Office	415	0.10	31.0	63.14	137.29	2,000.00

Note. This table shows descriptive statistics for POS transaction amounts per Transaction Location based on microdata from the ECB SPACE 2024 survey. The sample includes 76,294 transactions across 18 euro area countries and all values are expressed in euros.

Table A.5 Descriptive Statistics of Transaction Amounts per Country (in euros)

Country	Count	Minimum	Median	Mean	Standard Deviation	Maximum
France	8603	0.10	21.00	33.79	115.66	7845.00
Spain	8485	0.06	18.00	31.74	209.52	9046.00
Italy	8168	0.20	20.00	28.79	97.09	7000.00
Croatia	5024	0.01	15.68	24.60	42.66	1093.25
Slovakia	4902	0.20	20.00	28.06	55.59	2000.00
Portugal	4755	0.07	18.50	26.07	40.43	1000.00
Greece	4674	0.01	20.00	39.11	522.50	35000.00
Finland	4524	0.05	27.02	41.10	136.11	8000.00
Austria	4329	0.01	21.00	32.91	61.22	1600.00
Ireland	3946	0.35	20.00	34.71	114.80	5139.85
Belgium	3829	0.10	22.50	40.54	118.65	5000.00
Estonia	2527	0.01	19.00	27.83	59.12	2125.00
Cyprus	2391	0.08	20.00	40.56	166.31	4000.00
Lithuania	2177	0.39	17.40	38.93	442.58	20000.00
Slovenia	2073	0.40	15.50	26.07	53.36	1297.68
Latvia	2061	0.01	19.00	28.10	71.45	2000.00
Luxembourg	2011	0.01	30.00	52.47	85.41	2000.00
Malta	1815	0.50	21.00	35.35	177.20	7270.00

Note. This table shows descriptive statistics for POS transaction amounts per Country based on microdata from the ECB SPACE 2024 survey. The sample includes 76,294 transactions across 18 euro area countries and all values are expressed in euros.

Table A.6 Descriptive Statistics of Transaction Amounts per Age Group (in euros)

Age group	Count	Minimum	Median	Mean	Standard Deviation	Maximum
18–24	6,865	0.10	19.00	28.73	63.66	2,555.00
25–29	4,551	0.20	20.00	41.98	293.52	9,046.00
30–34	8,227	0.01	20.00	36.83	250.00	20,000.00
35–39	6,088	0.01	20.00	33.00	66.91	2,030.00
40–44	6,444	0.08	20.00	31.44	86.32	4,750.00
45–49	8,877	0.01	20.00	35.61	377.08	35,000.00
50–54	5,978	0.06	20.00	29.86	44.43	1,000.00
55–59	6,940	0.01	20.00	30.64	94.33	5,000.00
60–64	5,864	0.01	20.00	30.86	62.70	2,000.00
65–69	6,366	0.07	17.43	32.34	142.94	7,845.00
70–74	3,925	0.10	20.00	30.84	57.67	1,550.00
75+	6,169	0.10	23.00	33.63	160.24	8,000.00

Note. This table shows descriptive statistics for POS transaction amounts per Age Group based on microdata from the ECB SPACE 2024 survey. The sample includes 76,294 transactions across 18 euro area countries and all values are expressed in euros.

8.2 Appendix B: Results

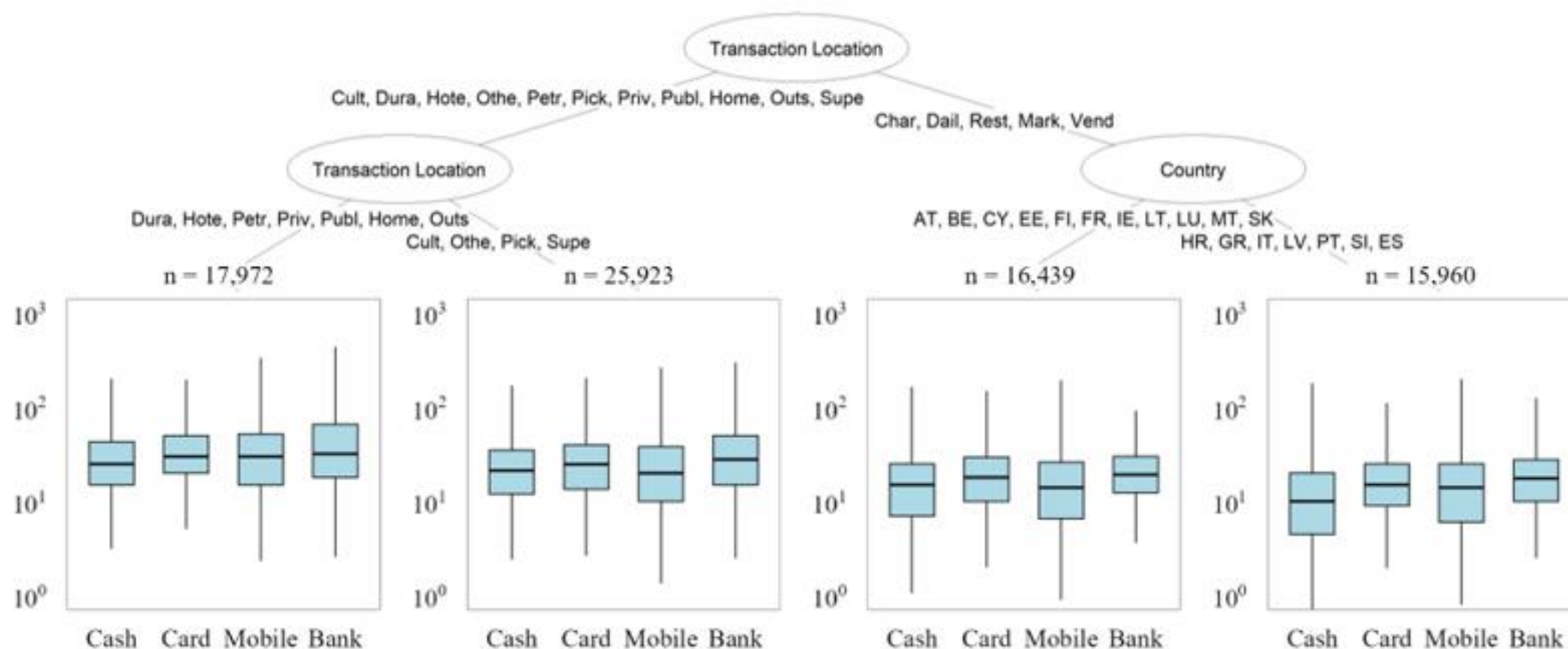


Figure B.1 MOB Tree Based on Full Control Set (All Variables)

Note. This model-based recursive partitioning (MOB) tree segments the data based on all available covariates of the SPACE data, including demographic, contextual, attitudinal, digital access, product ownership, and acceptance-related variables. Terminal nodes display boxplots of transaction amounts by payment method for each identified subgroup. The y-axis is plotted on a base-10 logarithmic scale, with a reduced display range from 10^0 to 10^3 (instead of the full range from 10^{-1} to 10^4) to improve clarity and focus on the main distribution. Country and transaction location labels are abbreviated; full names are available in Appendix Tables B1 and B2. For categorical splits, the left branch corresponds to the first listed category. Results are based on 76,294 POS transactions from the ECB SPACE 2024 dataset.

Table B.1: Abbreviations Used for Transaction Location Labels in MOB Plots

Full Label	Abbreviation
Culture/Sports/Entertainment	Cult
Durable Goods Shop	Dura
Hotel	Hote
Other Physical Location	Othe
Petrol Station	Petr
Pick-up Station	Pick
Private Payment	Priv
Public Authority/Post Office	Publ
Charity	Char
Day-to-Day Shops	Dail
Restaurant/Bar/Cafe	Rest
Street Vendor/Market	Mark
Vending Machine/Ticketing	Vend
Supermarket	Supe
Services Inside Home	Home
Services Outside Home	Outs

Note: This table displays the abbreviated labels used in the MOB regression trees and plots to simplify the display of transaction location categories.

Table B.2 Country ISO-2 Abbreviations Used in MOB Plots

Country	ISO-2 Code
Austria	AT
Belgium	BE
Croatia	HR
Cyprus	CY
Estonia	EE
Finland	FI
France	FR
Greece	GR
Ireland	IE
Italy	IT
Latvia	LV
Lithuania	LT
Luxembourg	LU
Malta	MT
Portugal	PT
Slovakia	SK
Slovenia	SI
Spain	ES

Note: This table displays the abbreviated labels used in the MOB regression trees and plots to simplify the display of country categories. Country labels have been abbreviated to their standard ISO-2 codes for display clarity.

Table B.3 Distribution of Transactions by Payment Method and Instrument Choice Availability

Payment Method	No Choice (%)	Choice Available (%)
Cash	20.9	79.1
Non-Cash	11.0	89.0

Note. This table reports the proportion of transactions conducted with and without payment choice, separately for cash and non-cash payments. Percentages are row-normalized and indicate, for each payment type, how often it was used in constrained versus free-choice contexts. The analysis is based on 76,294 POS transactions from the ECB SPACE 2024 dataset across 18 euro area countries.

8.3 Appendix C: Assumption Diagnostics

This appendix presents model diagnostics for all OLS regressions reported in Chapter 5, where the outcome variable is the log-transformed transaction amount and the main predictor is the payment method. Each figure below displays two panels: the left panel shows residuals plotted against fitted values, with a horizontal blue line at zero to assess linearity and homoskedasticity; the right panel shows a Q–Q plot comparing standardized residuals to a theoretical normal distribution, with the blue line indicating the expected quantile values under perfect normality. All regressions are based on POS transactions from the ECB SPACE 2024 dataset, covering 18 euro area countries.

Assumption diagnostics are presented in the order in which the models appear in the thesis and are grouped to minimize space.

Figure C.1 Residuals and Q–Q Plot (Hypothesis 1a – Ordinal – Model 1)

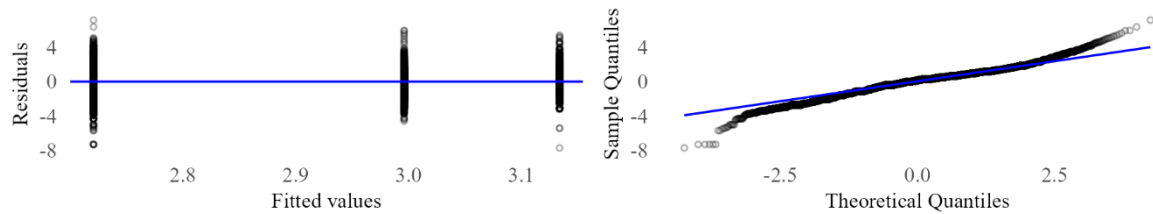


Figure C.2 Residuals and Q–Q Plot (Hypothesis 1a – Ordinal – Model 2)

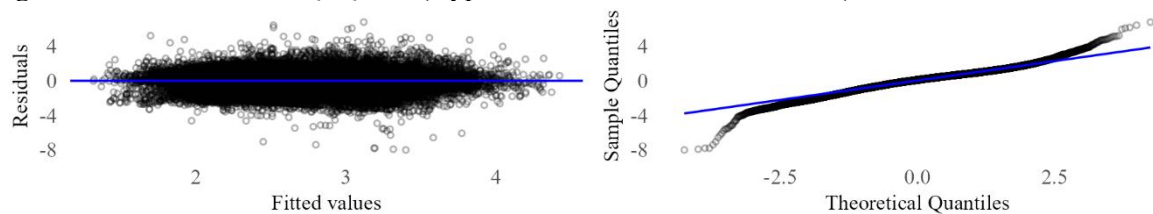


Figure C.3 Residuals and Q–Q Plot (Hypothesis 1a – Ordinal – Model 3)

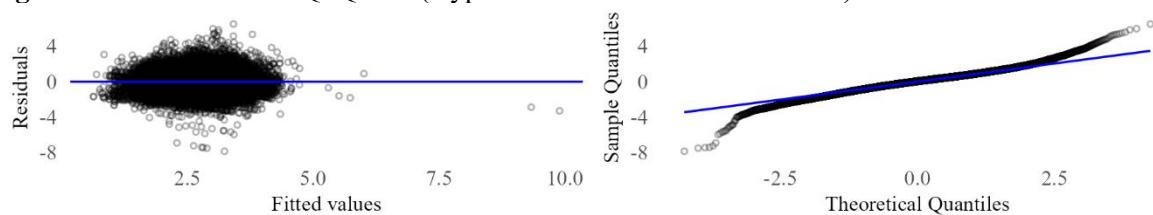


Figure C.4 Residuals and Q–Q Plot (Hypothesis 1a – Categorical – Model 1)

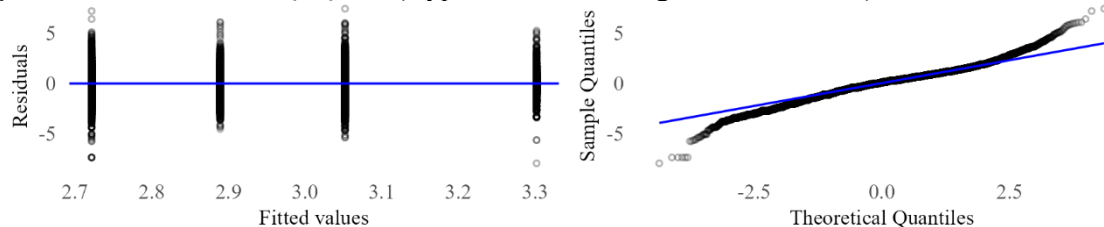


Figure C.5 Residuals and Q–Q Plot (Hypothesis 1a – Categorical – Model 2)

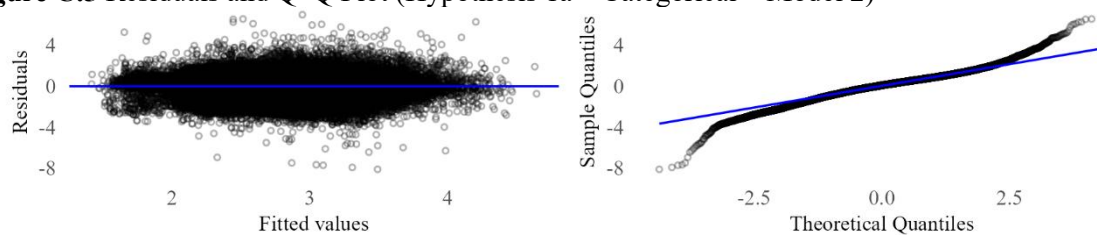


Figure C.6 Residuals and Q–Q Plot (Hypothesis 1a – Categorical – Model 3)

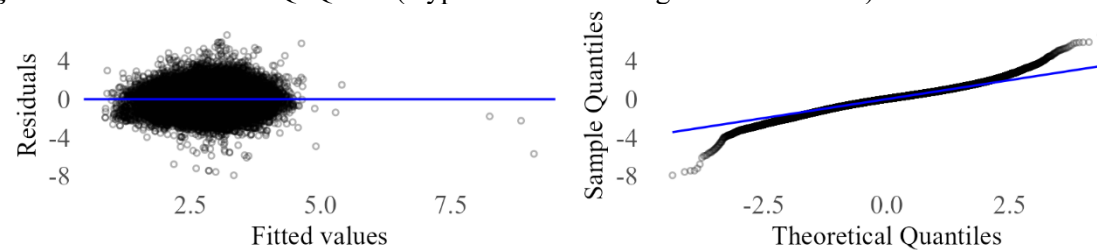


Figure C.7 Residuals and Q–Q Plot (MOB Transaction Location Group 1)

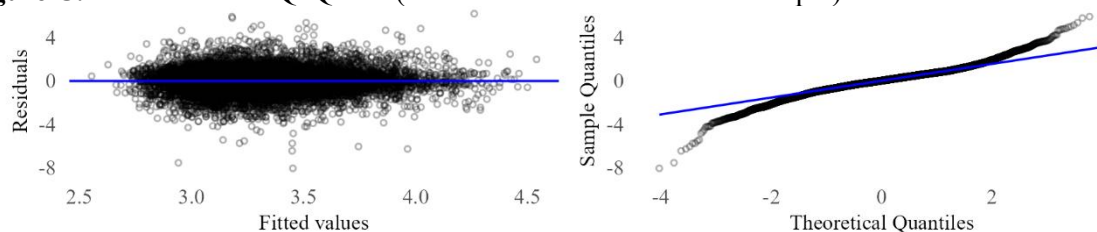


Figure C.8 Residuals and Q–Q Plot (MOB Transaction Location Group 2)

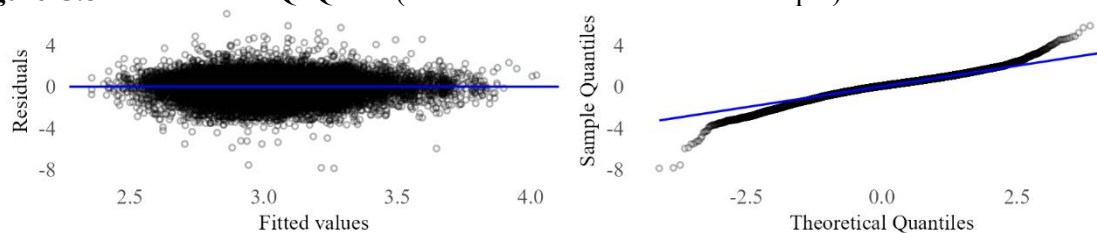


Figure C.9 Residuals and Q–Q Plot (MOB Transaction Location Group 3)

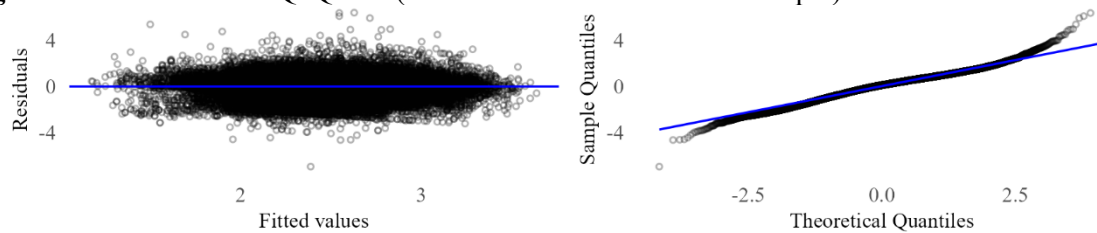


Figure C.10 Residuals and Q–Q Plot (Country Group 1)

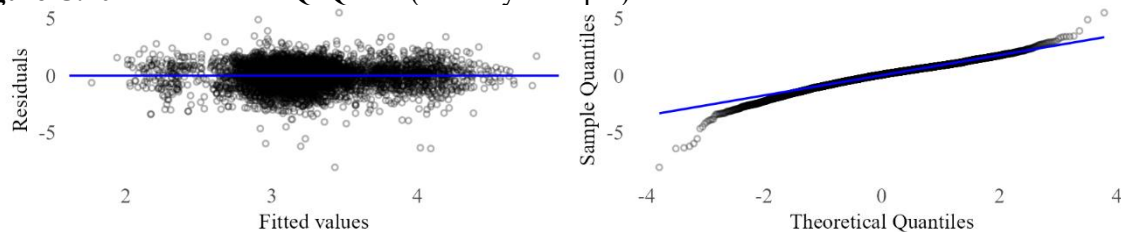


Figure C.11 Residuals and Q–Q Plot (Country Group 2)

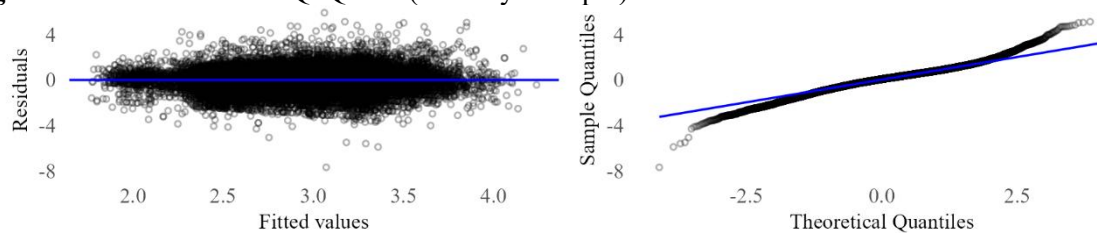


Figure C.12 Residuals and Q–Q Plot (Country Group 3)

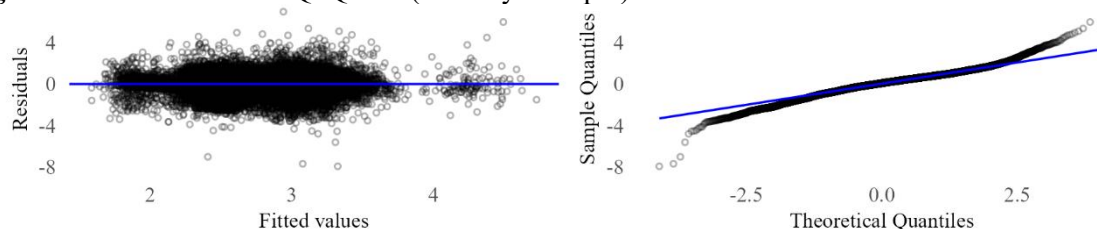


Figure C.13 Residuals and Q–Q Plot (Country Group 4)

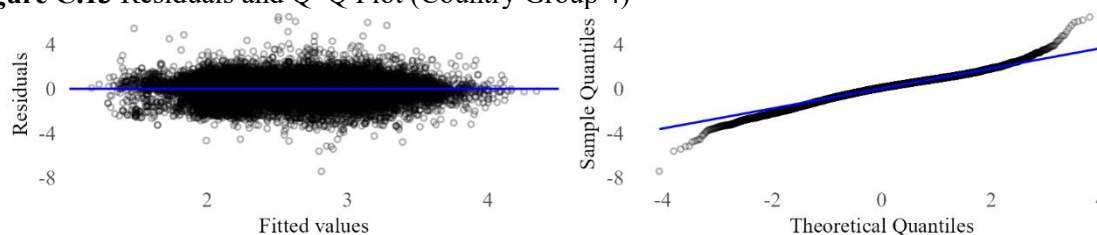


Figure C.14 Residuals and Q–Q Plot (Model 1 – Payment Choice Available)

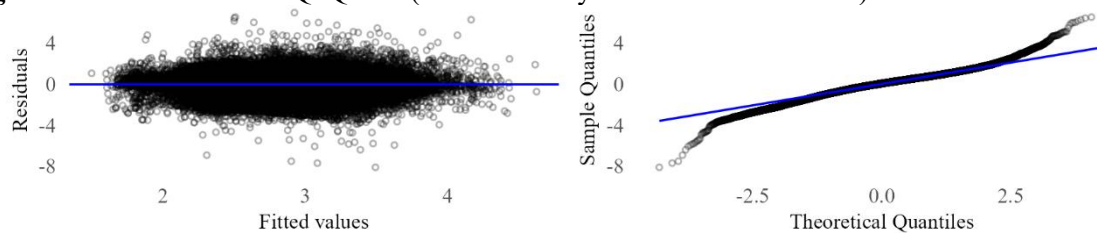


Figure C.15 Residuals and Q–Q Plot (Model 2 – No Payment Choice)

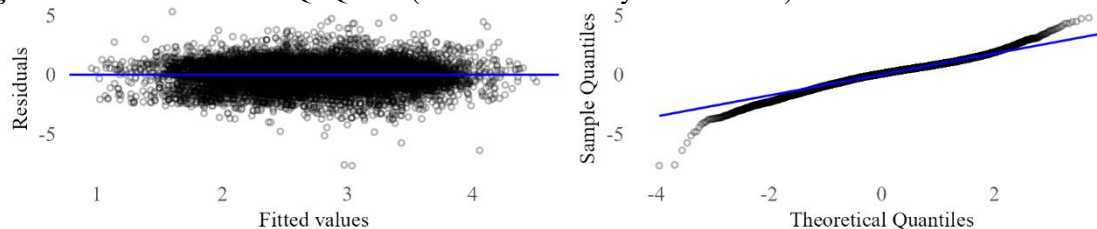
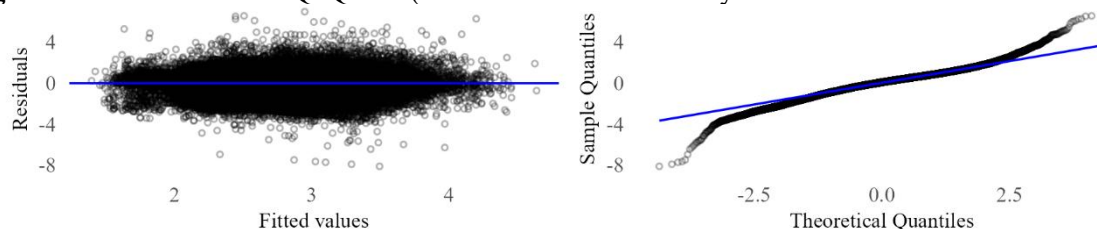


Figure C.16 Residuals and Q–Q Plot (Model 3 – Interaction: Payment Method $\times$ Choice Availability)



8.4 Appendix D: Variable List

Appendix D provides an overview of all variables retained for analysis in this thesis. Variables with more than 10% missing values were excluded, while those with less than 10% missingness were imputed using k-nearest neighbours ($k = 5$). The variable names used in this thesis are listed alongside their original survey codes, data types, and imputation status. The dataset is based on the ECB SPACE 2024 wave, which includes detailed information on consumer transactions and demographics across 18 euro area countries.

Variable Name	Original Code	Data Type and Categories	Imputed
ID	ID	Numeric	
Field	FIELD	Categorical: CATI, CAWI	
Survey_Month	MONTH	Categorical: April 2024, May 2024, June 2024, September 2023, October 2023, November 2023	
Day	DAY	Categorical: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday	
Country	COUNTRY	Categorical: Cyprus, Malta, Belgium, Italy, Spain, Greece, Austria, Estonia, Finland, Croatia, Ireland, Lithuania, Luxembourg, Latvia, Portugal, Slovenia, Slovakia, France	
Language	LANG	Categorical: Dutch, French, German, Estonian, Russian, English, Greek, Spanish, Italian, Latvian, Lithuanian, Luxembourgish, Maltese, Portuguese, Slovenian, Slovakian, Finnish, Swedish, Croatian	
DEM_Age	AGE	Categorical: 18–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, 70–74, 75+	
DEM_Education	EDUCATION	Categorical: Primary/lower secondary education, University/PhD/research, Upper/post-secondary education	KNN
DEM_Income	INCOME	Categorical: EUR 1,501–2,500, EUR 2,501–4,000, EUR 750 or less, EUR 751–1,500, More than EUR 4,000	KNN
DEM_HouseholdSize	HHSIZE	Ordinal: 1, 2, 3, 4, 5	KNN
DEM_Urban	URBAN	Categorical: Rural, Urban	
DEM_Gender	D1	Categorical: Female, Male, Other, non-binary	KNN
DEM_ActivityStatus	D6_1	Categorical: Employee, Self-employed, Without professional activity or student	KNN
Dem_Region	Derived	Categorical: Many groups of regions per country	KNN
Transaction_Amount	QA5A_8	Numerical	
Transaction_Location	QA6A_1	Categorical: Charity, Culture/Sports/Entertainment, Day-to-Day Shops, Durable Goods Shop, Hotel, Other Physical Location, Petrol Station, Pick-up Station, Private Payment, Public Authority/Post Office, Restaurant/Bar/Café, Services Inside Home, Services Outside Home, Street Vendor/Market, Supermarket, Vending Machine/Ticketing	KNN

Transaction_PaymentInstrument	Derived	Categorical: Cash, Card, Mobile, BankTransfer	
Transaction_InstrumentChoiceAvailable	Derived	Categorical: Yes, No	KNN
OWN_BankAccount	QQ1A_1	Categorical: Yes, No	
OWN_PaymentCard	QQ1A_2	Categorical: Yes, No	
OWN_NoPaymentMethod	QQ1A_4	Categorical: Yes, No	
OWN_OtherPaymentMethod	QQ1A_5	Categorical: Yes, No	
OWN_MobilePaymentDevice	QQ1A_6	Categorical: Yes, No	
OWN_Crypto	QQ1B_1	Categorical: Yes, No	KNN
OWN_Mortgage	D10_1	Categorical: Yes, No	
OWN_ConsumerLoan	D10_2	Categorical: Yes, No	
OWN_SavingsAccount	D10_3	Categorical: Yes, No	
OWN_OnlineBanking	D10_4	Categorical: Yes, No	
OWN_Investments	D10_6	Categorical: Yes, No	
OWN_NoProducts	D10_7	Categorical: Yes, No	
OWN_LifeInsurance	D10_9	Categorical: Yes, No	
OWN_PensionFund	D10_10	Categorical: Yes, No	
OWN_Smartphone	D9_1	Categorical: Yes, No	
OWN_Smartwatch	D9_2	Categorical: Yes, No	
OWN_NoSmartphoneOrSmartwatch	D9_3	Categorical: Yes, No	
INTERNET_Usage	D8	Categorical: About once a week, Every or almost every day, Less often, Never, No internet access, Two or three times a month, Two or three times a week	KNN
RECUR_Utilityies	QC1_2	Categorical: Yes, No	KNN
RECUR_Insurance	QC1_3	Categorical: Yes, No	KNN
RECUR_TelephoneInternet	QC1_4	Categorical: Yes, No	KNN
RECUR_Taxes	QC1_5	Categorical: Yes, No	KNN
RECUR_Subscriptions	QC1_6	Categorical: Yes, No	KNN
RECUR_Mortgage	QC1_7	Categorical: Yes, No	KNN
RECUR_SchoolFees	QC1_9	Categorical: Yes, No	KNN
RECUR_LoanRepayment	QC1_10	Categorical: Yes, No	KNN
RECUR_Rent	QC1_11	Categorical: Yes, No	KNN
CASH_StartingAmount	QA1	Numerical	KNN
CASH_WithdrawEase	QQ2	Categorical: Fairly difficult, Fairly easy, Very difficult, Very easy	KNN
CASH_WithdrawJourney	QQ2B	Categorical: Combined with commute, Combined with other activities, Never withdraw cash, Special journey	KNN
CASH_ATMFees	QQ11A	Categorical: Always pay fee, Most of the time, Never, Rarely, Sometimes	KNN
CASH_Reserves	QQ9	Categorical: Yes, No	KNN
PREF_CashVsCardGeneral	QQ3	Categorical: Card or other cashless payment, Cash, No clear preference	KNN
PREF_P2P_Payment	QQ17	Categorical: Bank transfers, Cash, Mobile or other online payments, No clear preference	
PREF_CashImportance	QQ4C	Categorical: Fairly important, Not important at all, Not so important, Very important	KNN
DIGI_Assistance	QQ15	Categorical: Yes, No	KNN
INCOME_CashShare	QQ8	Categorical: Between a quarter and a half, Between half and three-quarters, Half of your regular income is in cash, More than three quarters, None, Up to a quarter	KNN

INSTANT_Transfer	QQ1E	Categorical: Heard about it, but unsure, Never heard about it, No I can't, Yes I can, and I use the option, Yes I can, but I don't use the option	KNN
ADV_Cash_Acceptance	QQ13A_1	Categorical: Yes, No	
ADV_Cash_Faster	QQ13A_2	Categorical: Yes, No	
ADV_Cash_Privacy	QQ13A_3	Categorical: Yes, No	
ADV_Cash_Easier	QQ13A_4	Categorical: Yes, No	
ADV_Cash_Safer	QQ13A_5	Categorical: Yes, No	
ADV_Cash_Immediate	QQ13A_6	Categorical: Yes, No	
ADV_Cash_SpendingAwareness	QQ13A_7	Categorical: Yes, No	
ADV_Cash_Other	QQ13A_8	Categorical: Yes, No	
ADV_Cash_NoUse	QQ13A_9	Categorical: Yes, No	
ADV_Cash_None	QQ13A_10	Categorical: Yes, No	
ADV_Card_Acceptance	QQ13B_1	Categorical: Yes, No	
ADV_Card_Faster	QQ13B_2	Categorical: Yes, No	
ADV_Card_Easier	QQ13B_3	Categorical: Yes, No	
ADV_Card_Safer	QQ13B_4	Categorical: Yes, No	
ADV_Card_NoCashCarry	QQ13B_5	Categorical: Yes, No	
ADV_Card_SpendingAwareness	QQ13B_6	Categorical: Yes, No	
ADV_Card_Other	QQ13B_7	Categorical: Yes, No	
ADV_Card_None	QQ13B_8	Categorical: Yes, No	
ADV_Card_NoUse	QQ13B_9	Categorical: Yes, No	
PRIV_WishMorePrivacy	QQ16_1	Categorical: Yes, No	KNN
PRIV_DataConcern	QQ16_2	Categorical: Yes, No	KNN
PRIV_NoConcerns	QQ16_3	Categorical: Yes, No	KNN
PRIV_NoDigitalPayments	QQ16_4	Categorical: Yes, No	KNN
ACCEPT_CashNotAccepted	QA9_1	Categorical: Yes, No	
ACCEPT_DigitalNotAccepted	QA9_2	Categorical: Yes, No	
ACCEPT_DigitalAboveThreshold	QA9_3	Categorical: Yes, No	
ACCEPT_AlwaysAccepted	QA9_4	Categorical: Yes, No	
ACCEPT_NoPOSPayments	QA9_5	Categorical: Yes, No	
ISSUE_CardPayment	QA10_1	Categorical: Yes, No	KNN
ISSUE_MobilePayment	QA10_2	Categorical: Yes, No	KNN
ISSUE_CashRegister	QA10_3	Categorical: Yes, No	KNN
ISSUE_NoProblems	QA10_4	Categorical: Yes, No	KNN
NPS_EasierToUse	QQ18_1	Categorical: Yes, No	
NPS_Cheaper	QQ18_2	Categorical: Yes, No	
NPS_Security	QQ18_3	Categorical: Yes, No	
NPS_Privacy	QQ18_4	Categorical: Yes, No	
NPS_InnovativeFunctions	QQ18_5	Categorical: Yes, No	
NPS_Faster	QQ18_6	Categorical: Yes, No	
NPS_Incentives	QQ18_7	Categorical: Yes, No	
NPS_Recommendations	QQ18_8	Categorical: Yes, No	
NPS_OfflineUse	QQ18_9	Categorical: Yes, No	
NPS_NotInterested	QQ18_10	Categorical: Yes, No	
DuringDiary_CashWithdrawal	Q1_1	Categorical: Yes, No	KNN
DuringDiary_ReceivedCash	Q1_2	Categorical: Yes, No	KNN
DuringDiary_PaymentPOS	Q1_3	Categorical: Yes, No	KNN
DuringDiary_PickupPayment	Q1_4	Categorical: Yes, No	KNN
DuringDiary_CourierPayment	Q1_5	Categorical: Yes, No	KNN
DuringDiary_P2PPayment	Q1_6	Categorical: Yes, No	KNN
DuringDiary_OnlinePayment	Q1_7	Categorical: Yes, No	KNN
DuringDiary_HomeServicePayment	Q1_8	Categorical: Yes, No	KNN